An Executable Observer for Observer Patch Holography: Specification, Certified Components and Acceptance Criteria

Bernhard Mueller[1,2,*] Mario Poneder Jinwook Kim[3]

October 17, 2026

Paper release: r2050 **Released:** October 17, 2026

[1]Pragma Research Inc. [2]Information Physics Institute [3]Oraclizer Labs
[*]Corresponding author: bernhard@floatingpragma.ai

Abstract

Observer Patch Holography (OPH) seeks an executable realization of its observer algebra whose compatible inter-observer measurements reproduce the observed Universe, with quantum-field, classical and gravitational physics entering as controlled effective descriptions of those measurements. This paper states the executable construction as a technical contract. The specification consists of a twelve-port carrier with icosahedral incidence, a native local repair law that writes a coin and current record at every event, a finite observer algebra of preparations, measurements, retained records and record-only feedback, declared services, and two accepted closure inputs. We give acceptance criteria for the completion target and report the certified components. A two-event observer whose retained record selects its next repair moves a final occupation from 1/6 to 35/264. Under uniform seam attempts the repair channels with discarded records have the exact quantum spectral gap $(5-\sqrt{5})/60$, and complete classical records recover every initial load population and permit an optimal recovery of the unknown quantum input with explicitly computed fidelity below one. A binary detector converts the electromagnetic coupling computed from a closure input into a recorded bit that controls the next repair, and scalar-field detectors carry refinement, occupation and complete-output error bounds. In a common source, quantum electric flux, charged fermions and a scalar field share one register; for a declared preparation its recorded feedback response is certified nonzero at infinite oscillator occupation. A four-read continuation retains the disturbed field between measurements and joins source-compiled native repairs to active recovery of a logical qubit, including its reference entanglement. The quantum record laws compatible with the native repair form an exactly classified convex family, and a fixed count protocol reads the coherence that separates its members; no comparison data exist. On the refining spherical source, a common subsequence defines a regular local spatial form with spectral dimension two, an actual Gaussian density limit at finitely many scaled native times, admitted coherent waves with a finite comparison-space propagation bound, and continuous classical Gaussian wave readouts. A controlled repair protocol has a nonvanishing mean response at fixed small amplitude in this limit. A premise map records which theorem antecedents the construction supplies and which enter as inputs. The open connections are spatial refinement of the full interacting construction, responsive geometry, physical clock calibration and independent empirical comparison.

1 Introduction

Observer Patch Holography (OPH) treats observations as primary. The completion target of its construction programme is an executable realization of the OPH observer algebra whose compatible

inter-observer measurements reproduce the observed Universe. Reproduction is judged on observations. The joint statistics of those measurements, together with their responses to interventions, must admit the observed 3+1-dimensional quantum-field, classical and gravitational descriptions as controlled effective descriptions, each in its regime of validity. No fundamental field or spacetime is reconstructed first. Fields, particles and geometry enter as effective descriptions of what observers measure and jointly interpret, and the data structures of an implementation do not change that ontology [13, 18].

OPH starts from one hypothesis, stated in full in the basis document and the flagship synthesis [3, 2]: the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the structure reads itself into existence. Because nothing lies outside it, whatever observers recover and implement needs no further justification, nothing may be assumed that observers have not recovered, and agreement fixes what counts as shared; the carrier, the observer algebra, the repair law and the closure inputs P and N are justified by being recovered, by running, and by reproducing what observers measure. The executable specification is where the account enters the loop; the loop itself has no first step. Here P is the screen grain, the pixel ratio of one screen cell, and N a record capacity in nats; Section 2.6 states how the construction uses them.

The order of exposition enters at the specification, because it is concrete and because observers stand there, holding records and inferring. The order of justification puts the hypothesis first, since the hypothesis is what exempts the specification from external justification. This paper stays at the entry point. It treats the executable specification as a candidate and writes down the contract by which that candidate is judged. The contract states what is specified and what counts as completion, which components are certified and with which exact results, which theorem antecedents those components supply, and the open connections to the completion target.

Two companion bodies of work frame that contract. The flagship synthesis [2] gives the informal account of OPH and selected results. The theorem library holds conditional theorems on consensus and repair [5], observers and records [6], gauge structure [7], spacetime and Einstein dynamics [8], and screen microphysics [9]. The labels A1, A2 and A3 used there name theorem premises. A1 is the carrier, federation and response layer of the executable specification, A2 the agreement condition with its endogeneity clause, and A3 the least-informative state selection that the no-outside rule imposes. A theorem that uses them states an implication. Section 8 records, route by route, which of those antecedents the executable construction supplies on its own finite objects.

Section 2 states the specification, namely the carrier, the native repair law and the family of its quantum record laws, the finite observer algebra, the declared services, the closure inputs and the time parameters. Section 3 states the completion target as five obligations with their acceptance evidence. Sections 4 to 7 report the certified components, in turn the finite observer and its records, scalar field detectors, a common interacting source in which quantum electric flux, charged fermions and a neutral scalar share one register, and a fixed count protocol that reads the coherence separating the record laws. Section 8 gives the premise map together with four retained audit controls, Section 9 the open connections and Section 10 the reproducibility contract. Every number in the paper is copied from a retained receipt or from a model document of the construction [10], cited at a pinned revision of the public repository [1].

Results carry the epistemic classes of the flagship, summarized in Table 1. Executed programs in Sections 4 to 7 are finite constructions under declared premises; their analytic observation limits are identified separately. Actions, preparations, couplings, instruments, services and schedules are inputs of the specification. None of these construction results is a comparison with independent observations of nature. Individual results state only the boundary specific to them. The theorems and propositions below are finite theorems; the items inside them that quote a certified enclosure,

table or replayed count are certified finite computations; Section 7.3 is a diagnostic, and the count protocol of Section 7.1 is a prospectively fixed conditional test.

Table 1: Epistemic claim classes of the flagship synthesis [2], reproduced.

Class	**Meaning**
Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

Six claim records of the repository [1] state the scope of the certified components. Table 2 lists them with the sections that state their content.

2 The specification

2.1 The carrier and its load configurations

The specification has one carrier with twelve primitive ports. Its incidence is the oriented boundary of an icosahedron, with thirty seams and twenty faces; five seams meet at every port, the neighbours of each port form a five-cycle, and every seam lies on two oppositely oriented faces [9, 2]. Write A for the adjacency matrix of the seam graph and $L = 5I - A$ for its Laplacian. The spectrum of L is 0, $5 - \sqrt{5}$ with multiplicity three, 6 with multiplicity five and $5 + \sqrt{5}$ with multiplicity three, so its characteristic polynomial is $x(x-6)^5(x^2 - 10x + 20)^3$ [22, 63, 56].

A load configuration assigns a nonnegative integer x_p to each port p, with total load $\sum_p x_p = 2$. There are 78 configurations. The 66 with two distinct occupied ports form the binary shell; the other twelve are collisions $|2u\rangle$ with load two at one port u. We write $\mathcal{H}_{78} = \mathbb{C}^{78}$ for the quantum load space with basis $|x\rangle$, $\mathcal{H}_{66}$ for the span of the binary shell, and M_{78}, M_{66} for the corresponding full matrix algebras. For a binary configuration occupied at ports u and w we also write $|uw\rangle$ or $|\{u, w\}\rangle$.

Ports are numbered 0 to 11 by the frame vectors u_p: $(-1, \varphi, 0)$, $(1, \varphi, 0)$, $(-1, -\varphi, 0)$, $(1, -\varphi, 0)$, $(0, -1, \varphi)$, $(0, 1, \varphi)$, $(0, -1, -\varphi)$, $(0, 1, -\varphi)$, $(\varphi, 0, -1)$, $(\varphi, 0, 1)$, $(-\varphi, 0, -1)$, $(-\varphi, 0, 1)$, with $\varphi = (1 + \sqrt{5})/2$. The thirty seams are the pairs with $|u_a - u_b|^2 = 4$, each written (a, b) with $a < b$, and

Table 2: Certified components and the repository's claim records that state their scope. Each record names the evidence files that state the component's scope.

Component	Sections	Claim record
Two-event native observer, classification of record laws, quantum mixing of the repair channels, recovery from complete records, phase protocol and its source-executed implementation	2.3, 2.4, 4.1, 4.4, 4.5, 7.1, 7.2	`OPH-MODEL-NATIVE-OBSERVER-EXECUTION`
Correctable record capacity and compact controller	4.2	`OPH-MODEL-OBSERVER-RECORD-CAPACITY`
P-dependent binary detector and retrospective coupling audit	4.3, 7.3	`OPH-MODEL-P-RESPONSE-EXECUTION`
Memory certificate at observer-local ticks for the reciprocal exchange law	4.7	`OPH-MODEL-N4-LOCAL-TICK-CERTIFICATE`
Scalar detector, two-read continuation and two-site source apparatus	5.1, 5.2, 5.3	`OPH-MODEL-SCALAR-RESPONSE-EXECUTION`
Common interacting source, occupation and readout certificates, smooth continuation, compiled native attachment and encoded memory, observer readouts, history repair, geometric and field readouts	6.1 to 6.8, except Proposition 6.13	`OPH-MODEL-COMMON-SOURCE-EXECUTION`

that order is the orientation used by the repair law below [56, 57].

The theorem library labels A1 the layer formed by carriers, their federation through typed seams and coherent triple overlaps, the oriented spherical support, and the complete reversible port response [4]. The executed construction realizes one carrier of that layer, namely its ports, seams and faces, its load configurations and the repair events on its seams. Coherent quantum operations run through twelve-port source controls. A six-level processor carries twelve source generators, one for each port, and every executed gate is the exponential of a real combination of these generators, applied to two- and four-dimensional proper codes through declared load and export services [43, 34]. These are single-carrier protocols. Separate spherical-screen executions and their common continuum operator for field, decay and wave observations have explicit multi-carrier access and refinement contracts [11]; they do not identify a physical current for the complete response algebra or discharge the full A1 attachment. Section 8 records those physical antecedents.

2.2 Native repair and its quantum record laws

Fix an oriented seam $e = (a, b)$ and a fair placement coin $c \in \{0, 1\}$. The native repair law is

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(\mathbf{e}_b - \mathbf{e}_a), \tag{1}$$

where $\mathbf{e}_p$ is the unit load vector of port p and y is the repaired configuration. The two endpoint loads become the floor and the ceiling of their mean; the coin assigns the ceiling to a when $c = 1$; all other loads are unchanged.

Proposition 2.1 (The classical seam rule)**.** *Let a stochastic rule on one seam (i) conserve $x_a + x_b$, (ii) end at a configuration minimizing $x_a^2 + x_b^2$, (iii) break any remaining tie by a random choice independent of the loads, and (iv) be invariant under exchange of the two endpoints. Then it is* (1) *with a fair coin.*

Proof. By (i) and (ii) the outputs are the floor and the ceiling of the mean. An odd total leaves two outputs, and by (iii) endpoint a receives the ceiling with a constant probability q. Exchange of the endpoints maps q to $1 - q$, so (iv) gives $q = \frac{1}{2}$. □

Without (iii), the rule that leaves the extra unit with the endpoint holding more satisfies the other three requirements. It fixes every binary configuration, so the binary shell carries no motion and the records carry no geometry. Requirement (iii) applies the no-outside rule to the tie: nothing recovered prefers one endpoint.

The record is $r = (c, j)$, the coin and the signed current. Compatibility on a seam is specified independently of the update; it requires the two endpoint loads to differ by at most one. On the quantum load space the record defines the current instrument C with Kraus operators and record-discarded channel

$$L^e_{(c,j)} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \qquad \mathsf{C}_e(\rho) = \sum_r L^e_r \, \rho \, L^{e\dagger}_r. \tag{2}$$

Both maps are written in Kraus form [84]. The contrast-record variant E keeps $r = (c, d)$ instead, with Kraus operators $M^e_{(c,d)} = 2^{-1/2} \sum_{x:\, x_a - x_b = d} |y(x,c)\rangle\langle x|$ and record-discarded channel E_e. Uniform independent seam attempts form the baseline schedule; on diagonal binary states the repair is then the lazy symmetric exclusion process with two particles, with generator $\frac{1}{60}\sum_e (S_e - I)$ per attempt, S_e exchanging the endpoint loads [15, 22].

Proposition 2.2 (Native repair instrument)**.** *Let $e = (a, b)$ be an oriented seam of the twelve-port carrier, let the placement coin be fair, and let C_e and E_e act on $\mathcal{H}_{78}$. Then:*

(i) For every record value the branch map $x \mapsto y(x, c)$ is an injective translation of its support. Consequently $\sum_r L^{e\dagger}_r L^e_r = I$, and C_e is completely positive and trace preserving (CPTP), also when tensored with an untouched reference system. The same holds for E_e.

(ii) Every branch conserves the total load, leaves spectator ports unchanged, never increases $\sum_p x_p^2$, and maps the binary shell into itself.

(iii) The current labels satisfy $c \in \{0, 1\}$ and $j \in \{-1, 0, 1\}$; all six labels occur on $\mathcal{H}_{78}$ and four occur on the binary shell.

(iv) C and E induce the same classical transition on every configuration, and

$$\mathsf{E}_e^2 = \mathsf{E}_e, \qquad \mathsf{E}_e\mathsf{C}_e = \mathsf{C}_e\mathsf{E}_e = \mathsf{E}_e, \qquad \mathsf{C}_e^n = \mathsf{E}_e + 2^{1-n}(\mathsf{C}_e - \mathsf{E}_e) \quad (n \geq 1).$$

On the binary shell E_e is a trace-preserving conditional expectation.

(v) With the seam-total parity $p = (x_a + x_b) \bmod 2$ one has $d = 2j + (2c - 1)p$, and E equals C followed by a binary record of p and this relabelling, as an identity of complete unnormalized branches.

Proof sketch. The inverse of a fixed-current branch adds j at a and subtracts it at b; every basis vector has one branch for each coin, so each column of the joint record/source isometry has two entries $2^{-1/2}$ with disjoint supports [20]. For fixed contrast, the unchanged spectators and the endpoint sum reconstruct the input, which gives (i) for E. Items (ii) and (iv) are the single-seam relations of the candidate register [15]. Item (v) follows from $j = \lceil d/2 \rceil$ for $c = 0$ and $j = \lfloor d/2 \rfloor$ for $c = 1$; the parity-refined Kraus operator $\Pi_p L_{(c,j)}$ has exactly the contrast group as support [37]. The executable checks all 312 outputs of the observer's two seams against an independent floor and ceiling rule [20]. □

The current instrument is not a classical coarse-graining of the contrast instrument. Grouping the labels of E sums density operators and cannot restore the coherent Kraus sums of C. Both are members of one family of quantum record laws, which Section 2.3 classifies. The candidate register of the construction also records laws that couple the native current to private six-level response factors at the two seam endpoints [15]. One of them, a reciprocal coherent exchange named N4, enters the local-tick certificate of Section 4.7; the other components use the native law and its contrast variant.

2.3 Admissible record laws

The classical rule fixes the diagonal of a record law and leaves its coherences free. For the seam $e = (a, b)$ write $\kappa(x) = (x_a, x_b)$ for the seam key and $D = \mathbf{e}_b - \mathbf{e}_a$. A channel Φ on $\mathcal{H}_{78}$ is an admissible record law for e if it acts as the identity on the ten spectator ports, reproduces the classical repair distribution on every $|x\rangle\langle x|$, commutes with $U_t = \exp(i \sum_p t_p N_p)$ for every $t \in \mathbb{R}^{12}$, and is invariant under the carrier rotation that exchanges a and b with the current reversed. The third condition is the quantum form of local load conservation: every recorded branch transports one definite load.

Theorem 2.3 (Coherence tables)**.** *A channel Φ is an admissible record law if and only if*

$$\Phi(|x\rangle\langle x'|) = \sum_j G_j(\kappa(x), \kappa(x'))\, |x + jD\rangle\langle x' + jD|$$

for endpoint-symmetric positive semidefinite matrices G_j on the seam keys that can carry the current j, with $G_j(\kappa, \kappa) = \Pr(j \mid \kappa)$. On $\mathcal{H}_{78}$ the free entries are five multipliers: k between the empty and the full seam, g between the empty seam and a waiting single load, h between the full seam and a waiting single load, a real m between the two waiting orientations, and n between a moving single load and a moving collision. Positivity gives $|g|, |h|, |n| \le 2^{-1/2}$ and $|m| \le \frac{1}{2}$, and $k = 1$ forces $g = h$. Repairing one seam with (k, g, h, m, n) and then with (k', g', h', m', n') gives $(kk', gg', hh', mm', n\bar{h}')$. Exactly two admissible laws are idempotent, those with $g = h = m = n = 0$ and $k \in \{0, 1\}$, and every law with $k = 1$ converges under repetition to the idempotent law with $k = 1$.

Proof sketch. Covariance places $\Phi(|x\rangle\langle x'|)$ in the span of units $|y\rangle\langle y'|$ with $y - y' = x - x'$. Complete positivity confines nonzero Choi entries to pairs with nonzero diagonal, which the classical law makes $(x, x + jD)$, so both sides carry the same current and the Choi matrix is the direct sum of the G_j. Trace preservation and the classical law fix the diagonals, and spectator invariance makes each G_j a function of the seam keys. Under repetition every coherence keeps its seam key, so each multiplier applies again; idempotence requires $k, g, h, m \in \{0, 1\}$ and $n = n\bar{h}$, and the bounds leave $g = h = m = n = 0$. Exact arithmetic in $\mathbb{Q}(\sqrt{2})(i)$ checks the composition law on every pair of seam keys for 64 ordered pairs of tables and on all 6,084 matrix units for two pairs [14]. □

Four members stand out (Table 3). They form a ladder: C is C_0 followed by a diagonal readout of the coin, E is C followed by a readout of the seam parity as in Proposition 2.2(v), and T is C followed by a readout of the seam total. The completion of C in Proposition 2.2(iv) is the case $g = h = n = \frac{1}{2}$, $m = 0$ of the theorem, and E is the idempotent law with $k = 1$. A recorder of an endpoint's pre-repair load keeps the classical law and fails covariance, because one of its branches carries two currents. Each member corresponds to a principle: C_0 records only what conservation requires, C records the tie-break as an event, E is the coarsest law for which one repair completes the seam, and T records the full seam occupation. None of these principles follows from the theorem premises as stated. The construction executes C, whose records are exactly the classical event data,

and compares it with E. All members assign the same probabilities to every classical observation, and Section 7 reads g in one coherent event.

Table 3: Four distinguished admissible record laws [14].

Law	Each event records	Values per seam	k	$g = h$	Idempotent
C_0	the current	3	1	$2^{-1/2}$	no
C	coin and current	6	1	$1/2$	no
E	coin and pre-repair contrast	10	1	0	yes
T	coin and both endpoint loads	12	0	0	yes

2.4 The finite observer algebra

An observer of length n on the carrier consists of a preparation on $\mathcal{H}_{78}$, a record tape $R^{\otimes n}$, seam policies s_k that map committed records and classical randomness to seams or to supplied detector outcomes, the coherent writes W_{s_k}, optional supplied instruments whose committed bits may enter s_k, and a halting rule after n writes; a policy reads only committed records. Definition 2.4 is the instance $n = 2$, and Sections 4.3, 5 and 6 add declared instruments to it.

Definition 2.4 (Two-event native observer)**.** Let $R = \mathbb{C}^6$ carry one basis vector per current label (c, j). The observer carrier is

$$\mathcal{H}_{\mathrm{cap}} = \mathcal{H}_{78} \oplus (R \otimes \mathcal{H}_{78}) \oplus (R \otimes R \otimes \mathcal{H}_{78}),$$

with sectors of dimensions 78, 468 and 2808, total 3354; the sector index is the phase, the number of filled record slots.

(a) *Preparation.* The scored preparation is uniform on the 66 binary configurations; the full input family uses all 78.

(b) *Measurement and record.* At phase zero the coherent write $W_e = \sum_r |r\rangle \otimes L_r^e$ acts on the seam $e = (0, 1)$.

(c) *Record-only feedback.* At phase one the seam is $s(r) = (0, 5)$ if the stored current is positive and $(0, 1)$ otherwise, and $V_1(|r\rangle \otimes \psi) = |r\rangle \otimes W_{s(r)}\psi$.

(d) *Forecast.* Before the second event the controller reads its stored letter and issues a supplied forecast $q(r)$ of a positive second current.

(e) *Halting.* At phase two the observer acts as the identity.

The controller reads only the committed first record; it has no access to source loads or future outcomes.

Proposition 2.5 (One CPTP map with classical-prefix preservation)**.** *Let $V_0 = W_{(0,1)}$ and V_1 be extended by zero to the other sectors, giving A_0 and A_1, and let P_2 project onto the full sector. Then:*

(i) V_0 and V_1 are isometries on every nonfull sector, including unreachable combinations of letters and loads, and $T(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger + P_2\rho P_2$ is CPTP;

(ii) *on the inhabited subspace $K = \mathcal{H}_1 \oplus \mathcal{H}_2$ every first-record projector Q_r obeys $T^*(Q_r) = Q_r$, so a committed first record is reread with probability one;*

(iii) *for the blank input with occupations $(1, 1, 0, \ldots, 0)$, the first-record projector of the letter $(0, 0)$, extended by zero to phase zero, has expectation 0 before the tick and $1/2$ after it, so writing a record is a transition distinct from preserving it;*

(iv) *for the equal superposition of $\{0, 2\}$ and $\{1, 2\}$, the reduced-density entry of the first-record pair $((0, 1), (0, 0))$ equals $1/4$ after the first write and $1/8$ after the second while the classical diagonals are unchanged, so coherence between different record values need not survive feedback.*

Proof sketch. Orthogonal retained labels and $W_s^\dagger W_s = I$ give $V_0^\dagger V_0 = V_1^\dagger V_1 = I$, hence $A_0^\dagger A_0 + A_1^\dagger A_1 + P_2 = I$. Every subsequent map intertwines Q_r on K. In (iv) the working state attached to the entry has occupations $\{1, 2\}$, on which the isometries of $(0, 5)$ and $(0, 1)$ overlap by $1/2$ [20]. The checkpoint adapter verifies all $78 + 468 + 2808$ columns and both the preservation identity on the 3276-dimensional checkpoint space and the blank-sector control [43, 69]. □

Lemma 2.6 (Feedback requires classical record preservation)**.** *Let Λ be a CPTP map from $R \otimes W$ to $R \otimes W'$ with $\Lambda^*(B \otimes I) = B \otimes I$ for every B in the full matrix algebra of R. Then $\Lambda^*(I \otimes O) = I \otimes \Phi^*(O)$ for a unital completely positive map Φ^*, so changing only the record preparation of a product input changes no output statistic on W'.*

Proof sketch. The operators $B \otimes I$ lie in the multiplicative domain of the unital completely positive map Λ^*, because the Schwarz inequality is an equality for them [76]. Then $\Lambda^*(I \otimes O)$ commutes with every $B \otimes I$ and has the stated form [20]. □

Lemma 2.6 fixes the shape of the observer contract. A record that steers the working state is preserved as a classical record, with its projectors fixed, while coherence between different record values may change.

2.5 Declared services

Each component names the services it assumes; they belong to the specification and enter the resource counts where a component counts resources. The services are coherent proper-code load and export, with complementary failure outcomes retained by a complete channel definition; blank initialization, reset, storage and routing of quantum registers and classical records; Born readout of pointers and fair coins; and analogue precision of source pulses. Particular components add a calibrated external flux and probe (Section 4.3), incident-event recognition and bare-occupation readout (Section 4.7), and coherent phase preparation and readout (Section 7). Inside the original sharp joint instrument, native integer repair is a declared primitive. Section 6.6 instead attaches the compiled 78-dimensional C/E channels, routing and encoded recovery to the continuing work history. The processor algebra is M_6; its twelve generators are the exact source generators of the compiler, one per port [34, 43]; a proper code is a two- or four-dimensional subspace of $\mathbb{C}^6$ with its load and export channels and the complementary failure outcome; and the service error model is the compositional contract of Section 7.2. Exact certificates exclude numerical execution error and physical service error unless a component states otherwise.

2.6 The closure inputs P and N

The pixel ratio P is the area of one screen cell in units of $\ell_\star^2$, and N is a record capacity in nats [6, 28]. The construction accepts both as closure-derived inputs and fits neither to an output. P enters through a declared coupling map, the source-only structured-running branch of the fine-structure closure (one-loop running and matching conventions with the tree-level m_Z closure), defined in the P-derivation code of the repository [1] and consumed unchanged; the map is a declared selection, not a result of this construction. At the pinned closure value $P = 1.630972095858897\ldots$, the midpoint of the pinned forward-closure interval, the map gives $\alpha^{-1}(P) \approx 136.9948351774$ and $g(P) = \sqrt{4\pi\alpha(P)} \approx 0.302867613117$ in Heaviside–Lorentz natural units, numerical evaluations of the executed map [26, 65]. N enters only as a bound, and no component consumes its closure value. For a declared capacity carrier of dimension D, the record capacity $N_{\text{record}} = \ln D$ bounds every correctable classical record code by $\ln M_0 \leq N_{\text{record}}$ [13, 21]. The cosmological relation $N = 3\pi/(\Lambda\ell_\star^2)$ additionally requires a physical identification of capacity with horizon area, which the construction does not make.

2.7 Time parameters

The construction keeps the time parameters of Table 4 distinct [18]. Conversions between them need explicit maps. The only conversion it uses is the fixed-reference relation $t = -\beta s$ of Section 6.7 between a model coordinate and a modular parameter. No preferred universal physical clock is required.

Table 4: Time parameters and where they occur.

Parameter	**Meaning**	**Occurrence**
Host time	Runtime of a software execution; a resource cost	Never used as a physical quantity
Solver counter	Position in a search or repair schedule; the attempt count of a declared schedule	History repair (Section 6.8); attempt counts in the observer theorems
Causal order	Precedence of recorded events; counts of native operations	Observers, detectors, phase protocol, local ticks
Model coordinate	Parameter of a declared Hamiltonian or action	Scalar detectors, two-site apparatus, common source
Modular parameter	Parameter of the modular automorphisms of an algebra-state pair	Modular clock and fixed-reference relation (Section 6.7)
Operational clock	Observer-accessible reading with calibrated duration	Supplied by no component

3 Completion target and acceptance criteria

The completion target is an executable realization of the OPH observer algebra whose compatible inter-observer measurements reproduce the observed Universe. Their joint observations must admit a nontrivial 3+1-dimensional quantum field theory (QFT) as a controlled effective description, including the observed matter and interactions, and classical physics and dynamical gravity under appropriate coarse-graining. Four-dimensional QFT and 3+1-dimensional spacetime refer to the same dimension count of that description [13].

Success requires agreement of the accessible joint and conditional statistics and intervention responses in each description's regime of validity, with controlled errors and corrections. It does

not require every effective description to hold at arbitrarily fine scales. Meaning assignment connects actual preparations, instruments, interventions and records to shared physical quantities, and consensus and repair must preserve compatible joint and conditional statistics on overlapping observer domains. This is the proposed operational sense in which an objective reality emerges; agreement on an interpretation alone does not supply it [18]. Table 5 states the five obligations, their acceptance evidence, and the components of this paper that address each.

Rows two to four state no quantitative acceptance functional; an acceptance names the observables, the refinement family and the error budget before comparison. Exact equality of limiting observable laws is the mathematical target. Finite simulations and experiments are assessed with declared approximation and measurement errors. They cannot certify arbitrary empirical precision, and they do not force a particular random outcome in an independently repeated experiment. Agreement has to be constrained by measurements, because a consistent world model can describe physics different from that of nature.

Hardware, protocol and implementation choices are design variables, and no uniqueness theorem for a chosen rule is a prerequisite. Closure-derived P and N are accepted inputs under their declared contracts; accepting them discharges no other physical interface. Supplied background geometry, action, state, interpretation or clock data stay visible until the same construction accounts for them. The five obligations can advance in parallel. Passing the software checks verifies the retained contracts and leaves the physical target to the acceptance evidence above.

4 The finite observer

4.1 The native observer

The native observer executes Definition 2.4 on the scored preparation [19, 20]. A trial is one fresh preparation followed by two repair events and a halt; running many trials repeats the preparation and does not extend one observer's memory. Event order is the time parameter. The tape, prior, forecast table, seam selector and halting rule are designed apparatus, and the forecast is a fixed table without learning.

Proposition 4.1 (Statistics of the native observer)**.** *Prepare the 66 binary configurations uniformly, use fair coins and the controller of Definition 2.4, and let Y indicate a positive second current. Then:*

(i) the 264 histories, two coins for each of the 66 configurations at each of the two events, each have weight $1/264$*;*

(ii) the first record has masses $14/33$*,* $14/33$*,* $5/66$*,* $5/66$ *for* $(c, j) = (0, 0), (1, 0), (0, 1), (1, -1)$*, the conditional probabilities* $q(r) = \Pr(Y = 1 \mid r)$ *are* 0*,* $5/56$*,* 0*,* $1/2$ *in the same order, and* $\Pr(Y = 1) = 5/66$*;*

(iii) the Brier risk of forecasting Y is $395/7392$ *with* $q(r)$*,* $140/2013$ *with the best forecast based on the chosen seam alone, and* $12875/150304$ *when q is evaluated on an independent record resampled within the same seam;*

(iv) the mean final load at port 0 *is* $35/264$ *under record feedback and* $1/6$ *when the selector is disconnected or replaced by an independent selector with the same seam frequencies; the feedback effect is* $-3/88$*.*

Table 5: Completion obligations, acceptance evidence [13], and the corresponding evidence in this paper.

Obligation	**Acceptance evidence**	**This paper**
Executable observer algebra	Explicit state, operation, preparation, measurement, retained-record and feedback maps; their algebraic identities and observer-access limits; accounted memory and service assumptions; actual execution of the claimed protocols. A finite implementation discharges its own contract only.	**Discharged under the declared contract.** Two-event observers and detectors with every history retained (Sections 4, 5); a finite four-read continuation with compiled native repair and active encoded memory (Section 6.6). No federation or autonomous unbounded observer.
Shared operational meaning	Interpretation maps specified independently of the validation outcomes, compatible across observers and refinement, and preserving the declared intervention statistics and quantum correlations. Shared labels alone are insufficient; incompatible unperformed measurements need no common table of classical outcomes.	**Not discharged.** Committed-record preservation (Section 2.4); one record law for every observer of the common source; a schedule-independent completed history (Section 6.8). No interpretation maps across observers and refinement.
Effective quantum-field description	A common refining family of observer statistics and interpretation maps whose errors relative to a nontrivial 3+1-dimensional QFT tend to zero in the declared continuum domain, including the observed matter and interactions; the comparison concerns measurable correlations and intervention responses. Where an effective vacuum-sector mass gap is defined, it is nonnegative and may be zero.	**Not discharged.** Free-scalar refinement transfer (Section 5.1); two-site occupation limits with complete-output bounds and a nonzero interacting signal at infinite occupation (Sections 5.3, 6.2 to 6.5). No spatially refining interacting family.
Effective classical and gravitational descriptions	The same observer construction yields observations that admit compatible classical, geometric, stress and clock descriptions in their regimes, reproducing the gravitational predictions with controlled corrections. A fundamental metric or manifold is not a prerequisite; calculations on a supplied geometry need a map to these observer statistics.	**Not discharged.** A finite rank-three geometric readout, native causal cones and field relations (Section 6.8); modular clock readouts (Section 6.7). No stress, responsive geometry or calibrated clock.
Physical predictions	Explicit parameter and input ancestry, calibration and uncertainty; predictions and rejection rules fixed before independent comparison data; agreement with the observed statistics, including alternatives and distinguishing tests. Reusing a calibration datum is not an independent prediction.	**Decision rule fixed; no data.** A fixed count protocol that reads the record-law coherence g, without data (Sections 7.1, 7.2); a retrospective coupling audit (Section 7.3). The coupling map $P \mapsto \alpha(P)$ is a supplied input.

Proof sketch. For (iii), $\mathbb{E}[(Y - q(R))^2] = \mathbb{E}[\mathrm{Var}(Y \mid R)]$. The chosen seam S is a function of R; the best S-only forecast has excess risk $\mathbb{E}[\mathrm{Var}(q(R) \mid S)]$, and an independent resample within the seam adds twice that amount. For (iv), a positive first current occurs on ten binary inputs with coin zero, mass $5/66$. On that branch $x_0 = 0$, $x_1 = 1$ and $\mathbb{E}[x_5] = 1/10$; a second repair on $(0, 1)$ gives port-0 mean $1/2$ and one on $(0, 5)$ gives $1/20$. Hence the change is $(5/66)(1/20 - 1/2) = -3/88$ and the mean is $1/6 - 3/88 = 35/264$. An independent selector leaves the uniform shell uniform under either seam [20]. □

The other two current letters occur only on the full carrier, where the forecast abstains. The exact receipt retains all 264 histories and 18 calculation checks [60]. A seeded sequential execution of 20,000 fresh trials (seed 20261012) performs 40,000 native events, matches all 20,000 trials against their exact path references and records 20,000 explicit halts [61]. Its sample means of the final port-0 load are 0.1327 with feedback, 0.16645 with the selector disconnected and 0.1668 with the independent selector; the sample Brier risk of the record forecast is 478659/8960000, against 395/7392.

4.2 Record capacity

The retained records are evaluated with the correctable-public-record construction of the theorem library [6, 21]. There M_0 is the largest correctable set of reachable public records, the independence number of the compound checkpoint confusability graph; the record capacity is $N_{\mathrm{record}} = \ln M_0$ nats; and $M_0 \le D$ for a declared carrier of dimension D. The counts below describe the reachable record interface; program, source, probe and service memory are additional.

Proposition 4.2 (Record capacity of the native observer)**.** *Take halt and exact classical reread as the declared terminal continuations of the observer of Definition 2.4. Then the compound graph has no edges, and:*

(i) $M_0 = 11$ *on the binary preparation (four reachable first records, 264 histories) and* $M_0 = 17$ *on the full family of 78 inputs (six first records, 312 histories);*

(ii) each terminal word projector has rank 78 in the 3354-dimensional carrier, distinct projectors are orthogonal, and the eleven reachable ones have total rank 858, so the bound is unsaturated with slack $\ln(3354/11)$*;*

(iii) the binary preparation reaches $1+4+11 = 16$ *classical prefixes, so four bits encode the complete controller history; a compact controller that reads only this code reproduces the phase, action, forecast, source loads, record and halt of all 264 histories; the full family reaches 24 prefixes and needs five bits;*

(iv) coarser retained readouts give correctable codes of size 4 for the first letter alone, 4 for the second letter alone, 3 for the forecast alone, 2 for the chosen seam alone, 4 for forecast and seam, and 1 after complete erasure.

Proof sketch. Halt and reread leave the tape unchanged, so the joint kernel is the identity and the existing evaluator returns the number of reachable terminal words. The terminal projector of word w occupies the basis coordinates $546 + 78w$ to $546 + 78w + 77$ of the carrier [21, 62]. □

The terminal capacity is $\ln 11$ nats, about 3.459 bits, on the binary preparation and $\ln 17$ nats, about 4.087 bits, on the full family. The Shannon entropy of the benchmark record law is a different quantity, about 2.023 nats. The four-bit code is the minimum fixed-width encoding of the

16 reached prefixes; a naive layout with a separate phase uses eight bits, and an optimal index over all 43 allocated prefixes uses six.

The library's global proposal adds closure across the complete terminal fiber, $N_\star = \ln D_\star = \ln M_0(U_{N_\star})$, and identifies the same carrier with a de Sitter horizon through $N_\star = A_{\mathrm{dS}}/(4\ell_\star^2)$ and $A_{\mathrm{dS}} = 12\pi/\Lambda$, hence $\Lambda\ell_\star^2 = 3\pi/N_\star$ [6, 21]. The native records supply an actual finite input to that evaluator. The complete-fiber closure and the horizon identification are separate steps that this construction does not take.

4.3 The P-dependent detector

The detector carries the closure input P through the electromagnetic coupling into a recorded binary observation that controls the next repair [26, 27]. The charge assignment, probe, calibrated flux, readout and controller are specified apparatus, and the coupling branch keeps its documented endpoint status (Section 7.3).

Choose the charged realization $Q_v = g(P)N_v$ of the native occupation N_v. A current record j on the seam (a, b) then transfers the charge $\Delta Q = g(P)\, j\, (\mathbf{e}_b - \mathbf{e}_a)$, and total charge is conserved. For port gauge parameters λ_v and $U_\lambda|x\rangle = e^{ig\sum_v \lambda_v x_v}|x\rangle$, every current-grouped Kraus operator obeys $U_\lambda L_{(c,j)} U_\lambda^\dagger = e^{igj(\lambda_b-\lambda_a)} L_{(c,j)}$. The probe is a unit-charge path qubit prepared in $|{+}X\rangle$. The first native current routes it around a calibrated external loop with signed exposure $j\chi$, where $\chi = \oint A \cdot dx$ is prepared independently of g, so the relative phase is $\varphi_j = g(P)\, j\, \chi$, the Aharonov-Bohm phase [72]. A Y measurement commits $y = \pm 1$ with amplitudes $m_y(j) = \frac{1}{2}[e^{-i\varphi_j/2} - iy\, e^{i\varphi_j/2}]$ and effects $E_y = \sum_r \frac{1}{2}[1 + y \sin\varphi_{j(r)}]|r\rangle\langle r|$ on the old current register. The actuator receives only y. It repairs $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, keeps y unchanged, and the protocol halts.

Proposition 4.3 (P-dependent response)**.** *Prepare the 66 binary configurations uniformly, run the protocol above, and put $s = \sin(g(P)\chi)$. Then:*

(i) $\{E_y\}$ *is a binary instrument on every input, including coherent records; each* $M_y = \sum_r m_y(j(r))|r\rangle\langle r|$ *commutes with the old record projectors, and the composed experiment is one CPTP map that preserves the committed bit;*

(ii) $\Pr(y \mid j_1, P, \chi) = \frac{1}{2}[1 + y \sin(g(P) j_1 \chi)]$*;*

(iii) at $\chi = 1$ *all 528 state, coin, detector and coin histories have positive probability, and*

$$\mathbb{E}[y] = 0, \quad \mathbb{E}[y j_1] = \tfrac{5}{33} s, \quad \mathbb{E}[j_2] = \tfrac{5}{264} s, \quad \mathbb{E}[g(P) j_2] = \tfrac{5g(P)}{264} s;$$

(iv) for weak exposure, $\mathbb{E}[g\, j_2] = \frac{5\pi}{66}\alpha(P)\chi + O(\chi^3)$*;*

(v) disconnecting the probe, or replacing y by an independent fair bit, gives $\mathbb{E}[j_2] = 0$ *with the same detector and seam frequencies.*

Proof sketch. The eigenvalues of E_y lie in $[0, 1]$ and sum to one, which gives (i). For (iii), after the first repair let the source be z; averaging the second coin gives $\mathbb{E}[j_2 \mid z, j] = \frac{1}{4}(2z_0 - z_1 - z_5) + \frac{sj}{4}(z_1 - z_5)$. The first term averages to zero because the first repair preserves the uniform shell. Currents ± 1 each have mass $5/66$, and on either branch the other load is uniform over ten ports, so $\mathbb{E}[j(z_1 - z_5)] = 5/66$. An independent floor and ceiling enumerator reproduces every coefficient over all $66 \cdot 2 \cdot 2 \cdot 2 = 528$ paths [27, 29]. □

At the computed P and $\chi = 1$ the plus probability is 0.649129261647 for $j = +1$, 0.350870738353 for $j = -1$ and exactly $1/2$ for $j = 0$. The mean second current is 0.00564883566843, the mean transferred charge 0.00171084937579 in natural units, and the equivalent source audit gives a final port-0 load of 0.161017830998235 against $1/6$ for the controls. Off-closure parameter controls at $0.999P$ and $1.001P$ recompute the coupling map and change the detector and feedback probabilities at fixed χ [26, 65]. Fifteen calculation checks in the receipt, six independent tests and a fixed-seed sequential run of 200,000 trials (seed 20261012) support the construction; the run's mean second current is 0.006390 with standard error 0.00085715, consistent with the analytic value.

The actuator's record algebra is $\mathbb{C}(\{-1, +1\})$, with correctable size $M_0 = 2$. The scientific audit retains (r_1, y, r_2), with 24 terminal words, 37 prefixes, a six-bit encoding and $M_0 = 24$ [28]. If an identified cut enclosing a retained record obeys the area-capacity law with area equal to K chart cells, then $N_{\mathrm{cut}} = KP/4$ and $\ln M_0 \le N_{\mathrm{cut}}$ give $K \ge 4 \ln M_0 / P$. With the pinned P, $4 \ln 2/P \approx 1.6999608573$ and $4 \ln 24/P \approx 7.7942567832$, so one retained detector bit needs $K \ge 2$ and the full audit $K \ge 8$ [28].

4.4 Quantum mixing and memory

Discarding every seam, coin and repair record after each event turns the repairs into channels Φ_{C} and Φ_{E} on the source. Under independent seam attempts these channels have an exact spectral theory [22]. Theorems 4.4 and 4.5 concern this record-discarded channel under independent uniform attempts; they do not describe the retained tape or the feedback controller.

Theorem 4.4 (Signed-difference reduction and quantum gap)**.** *Let G be a finite connected graph with $n \ge 2$ vertices, positive edge weights w_{ab} and $W = \sum w_{ab}$. Let each event choose the edge ab with probability w_{ab}/W and a fresh fair coin, and discard its record. On the binary space $\mathcal{H}_k$ of load k, $1 \le k \le n-1$, with $D = \binom{n}{k}$:*

(i) for a matrix unit $|x\rangle\langle y|$ the signed difference $z = x - y$ is invariant; if z has r positive entries and $Z = \{v : z_v = 0\}$, then on the sector of z

$$\Phi_{\mathsf{L}}\big|_z = I - \frac{L_{\mathrm{excl}}(G[Z], k - r) + K_{\mathsf{L}}(z)\, I}{2W}, \qquad \mathsf{L} \in \{\mathsf{C}, \mathsf{E}\},$$

with the exclusion Laplacian L_{excl} of the induced graph, $K_{\mathsf{C}}(z) = \sum w_{ab}|z_a - z_b|$ and $K_{\mathsf{E}}(z) = 2\sum w_{ab}\mathbf{1}[z_a \ne z_b]$;

(ii) both channels are unital and Hilbert-Schmidt self-adjoint with spectrum in $[0, 1]$, their fixed algebra consists of the scalars, and both have gap $\gamma = \lambda_1(G)/(2W)$, where $\lambda_1(G)$ is the first positive Laplacian eigenvalue; and $\|\Phi_{\mathsf{L}}^t - \mathrm{Dep}_D\|_{2\to 2} = (1 - \gamma)^t$;

(iii) for every input and every untouched reference R, $\mathrm{dist}\big((\Phi_{\mathsf{L}}^t \otimes \mathrm{id}_R)(\rho), \tau \otimes \rho_R\big) \le \min\{1, \frac{D}{2}(1 - \gamma)^t\}$, with $\tau = I_D/D$ and $\mathrm{dist}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.

For the twelve-port carrier at load two ($n = 12$, $W = 30$, $\lambda_1 = 5 - \sqrt{5}$, $D = 66$) the gap is $\gamma = (5 - \sqrt{5})/60 \approx 0.046066$, and the slowest off-diagonal sectors contract by exactly $r_{\mathsf{C}} = 5/6$ and $r_{\mathsf{E}} = 7/10$ per attempt. On the full 78-state carrier the gap is the same, the full channel is block triangular with a diagonal collision block whose entries lie in $[0, 5/6]$, and the unique stationary density is the uniform binary density, of rank 66 on dimension 78.

Proof sketch. A matrix unit survives a coin branch exactly when its two records agree, and then both configurations translate by the same current; comparing the coin branches gives (i). The diagonal

sector is the symmetric k-particle exclusion chain, whose gap equals $\lambda_1(G)$ by the theorem of Caputo, Liggett and Richthammer [75]. For a nonzero sector, $K_{\mathsf{E}} \geq K_{\mathsf{C}} \geq \frac{1}{2}\sum w_{ab}(z_a - z_b)^2 \geq r\lambda_1 \geq \lambda_1$, so no off-diagonal mode is slower. Item (iii) uses Schmidt rank at most D and $\|Y\|_1 \leq D\|Y\|_2$. On the carrier, $r = 1$ gives $K_{\mathsf{C}} = 10$ and $K_{\mathsf{E}} = 2(10 - \mathbf{1}[uv \text{ edge}])$, while $r = 2$ gives $K_{\mathsf{C}} \geq 16$ and $K_{\mathsf{E}} \geq 24$. The collision block has largest entry $1 - d_{\min}/W = 5/6 \leq 1 - \gamma$ [22]. The executable checks the reduction against the native primitive on all 4,356 binary matrix units and the transient structure on all 6,084 full matrix units [63]. □

Choose adjacent ports u, v and the two mixtures $\rho_\pm = \frac{1}{20}\sum_{w\notin\{u,v\}}(|uw\rangle \pm |vw\rangle)(\langle uw| \pm \langle vw|)$. They have identical load populations and orthogonal supports, and

$$\operatorname{dist}(\Phi_{\mathsf{L}}^t(\rho_+), \Phi_{\mathsf{L}}^t(\rho_-)) = r_{\mathsf{L}}^t, \qquad \operatorname{dist}(\Phi_{\mathsf{C}}^t(\rho_+), \Phi_{\mathsf{E}}^t(\rho_+)) = \tfrac{1}{2}(r_{\mathsf{C}}^t - r_{\mathsf{E}}^t).$$

The second distance is $1/15$ after one attempt and peaks at four attempts at $12259/101250$; the binary effect $\frac{1}{2}(I + A_{uv} + A_{uv}^\dagger)$ with $A_{uv} = \sum_{w\notin\{u,v\}}|uw\rangle\langle vw|$ attains it. Load populations, classical load histories and diagonal geometric observables obey the same exclusion process under both laws, so only coherent preparations and readouts distinguish them under this schedule.

Theorem 4.5 (Finite quantum memory)**.** *Under the schedule of Theorem 4.4 on the twelve-port carrier:*

(i) on the binary shell both record-discarded channels are entanglement breaking after at most 134 attempts;

(ii) on the full carrier let u, v be adjacent and $|\Psi\rangle = (|2u\rangle|0\rangle + |uv\rangle|1\rangle)/\sqrt{2}$ with a reference qubit; after t attempts the coefficient of $|2u\rangle\langle uv|$ is a_{L}^t, with $a_{\mathsf{C}} = 23/30$ and $a_{\mathsf{E}} = 7/10$, and the partial transpose on the reference has a principal 2×2 minor equal to $-a_{\mathsf{L}}^{2t}/4 < 0$; no finite power of either full channel is entanglement breaking;

(iii) every input converges to $\tau_P \otimes \rho_R$, where τ_P is the uniform binary density, with dist $\leq \min\{1, \mu^{s/2} + \mu^s + \frac{D}{2}(1-\gamma)^{t-s}\}$ *for all $0 \leq s \leq t$, $\mu = 5/6$ and $D = 66$.*

Proof sketch. For the normalized Choi state J_t, $\|J_t - I/D^2\|_2^2 = D^{-2}\sum_{i>0}\lambda_i^{2t}$. The separable ball of Gurvits and Barnum [80] and the equivalence of separable Choi states with entanglement-breaking channels [82] reduce (i) to $65\, r_{\mathrm{mix}}^{2t} + 4290\, r_{\mathsf{L}}^{2t} \leq 1/4355$, where $r_{\mathrm{mix}} = 1 - \gamma$. The rational enclosure $559/250 < \sqrt{5} < 2237/1000$ certifies this at $t = 134$ and its failure at $t = 133$; treating all modes with r_{mix} would certify only 178. For (ii), 25 edges avoid u and four of them touch v; C keeps one coin on each of those four and E keeps neither, and the triangular structure prevents any return from the binary space. Separable states have positive partial transpose [86]; the complete argument is in [22, 63]. □

Replacing the full carrier by its attracting binary shell therefore changes an exact finite-time quantum property, although both converge to the same stationary density. The number 134 is a sufficient certificate time; the exact index of entanglement breaking on the shell is not determined.

4.5 Recovery from complete records

Preserving committed records and recovering the unknown input from them are different questions [23]. Here a seam policy may depend on committed records and classical randomness and may not inspect hidden loads or insert quantum corrections; a final decoder may act on the full load register and tape, keep the tape, use blank ancillas and return a state on the original carrier. That decoder access is stronger than the record-only access of the observer's controller.

Theorem 4.6 (Optimal recovery from complete native records)**.** *Let X be the 78 configurations or the 66-state shell and $D = |X|$. For a complete admitted history h let X_h be its compatible inputs, P_h the projector onto them, $d_h = \operatorname{rank} P_h$, and w_h the product of its seam and coin factors. Then:*

(i) the history Kraus operator is $K_h = \sqrt{w_h}\, U_h P_h$ with $U_h^\dagger U_h = P_h$ and $\sum_h w_h P_h = I$; a tape-controlled inverse outputs $\sum_h w_h |h\rangle\langle h| \otimes P_h \rho P_h$, so every initial load population is recovered;

(ii) among all final CPTP decoders the optimal entanglement fidelity for the maximally mixed input is $F_{\mathrm{opt}} = D^{-2} \sum_h w_h d_h^2 = D^{-2} \sum_{x,y} H_{xy}$, with $H_{xy} = \sum_{h:\, x,y \in X_h} w_h$;

(iii) for the two-event observer, F_{opt} equals 5543/8712 (binary) and 1301/2028 (full) under C, and 769/1452 and 3329/6084 under E with the same current-sign selector applied to its decoded current;

(iv) the maximal exactly correctable algebra on the whole input is the commutant of $\{P_h\}$, and one event with full-support random seam choice reduces it to the diagonal load algebra; nevertheless the qubit code $|0_L\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $|1_L\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ satisfies the Knill-Laflamme condition $QP_hQ = \alpha_h Q$ for every seam, coin and record of either law and is recovered exactly after one event;

(v) under independent seam attempts $H_{xy}(t) = q_{\mathsf{L}}(x-y)^t$, with largest off-diagonal rates 5/6 for C and 7/10 for E on both carriers; $F_{\mathrm{opt}}(t) \to 1/D$, and neither retained instrument is entanglement breaking at any finite horizon;

(vi) a fixed E seam word identifies every full-carrier input exactly when at most one visited component has size one or two, and every binary input exactly when at most one component is a singleton and at most one has size two; the minimal such words have eight and seven distinct edges;

(vii) no bounded record-only adaptive policy makes C entanglement breaking; a C word that doubles a minimal forest, 16 or 14 events, excludes every nontrivial exact code; the policy that repeats each forest edge until its coin first differs identifies the input almost surely, with stopping time $T = m + \sum_{i=1}^{m} G_i$ for independent geometric G_i of success probability one half, $\Pr(T = t) = \binom{t-m-1}{m-1} 2^{-(t-m)}$ for $t \geq 2m$, and $\mathbb{E}[T] = 3m$, namely 24 on the full carrier and 21 on the binary shell.

Proof sketch. Every retained current determines a translation, so reversing them gives the input uniquely from the terminal configuration and h. Optimality follows from Cauchy-Schwarz on each classical branch: $\sum_a |\mathrm{Tr}(R_{ha} U_h P_h)|^2 \leq d_h^2$. For (iv), commutation is sufficient by (i), and necessity follows from equality in the Schwarz inequalities in a Stinespring dilation; the code has nonadjacent occupied pairs, equal one-port means and vanishing products $N_a N_b$ on every edge, which gives the Knill-Laflamme identity [83]. For (vi), matching E histories are those for which z is constant on every visited component. For (vii), the inputs $\{u, w\}$ and $\{v, w\}$ with u, v nonadjacent keep equal currents for at least one coin on every edge, so a history of length at most T shares a rank-two fiber with probability at least 2^{-T}; repeating an edge with the opposite coin reveals the initial parity through $d = 2j_1 + j_*$ [23, 64]. □

The minimal fixed E words are the forests $(0,1),(0,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the full carrier and $(1,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the binary shell, with component sizes $(3,3,3,3)$ and $(1,2,3,3,3)$. The full-input correctable blocks of the two-event observer

have sizes $46, 10, 9, 1$ on the binary shell and $56, 10, 9, 1, 1, 1$ on the full carrier, for both laws. Existing source inverse words implement the single-seam branch reversal of (i) while keeping the classical archive [23, 64]. Native current records thus suffice to reverse classical load transport, without restoring arbitrary coherence lost across record groups. The source encoder and active recovery of Section 6.6 instead preserve the stated logical qubit between events; they do not invert the whole bare-load channel.

4.6 A coherent continuation and its selection boundary

An exact finite experiment attaches a reflection walk to the binary native branches, with coherent event access in place of a committed tape [24]. For each binary configuration x, seam e and coin c, let (y, j) be the native output and current. On the $66 \cdot 60 = 3960$ arc basis vectors define

$$S|x, e, c\rangle = |y, e, c'\rangle, \qquad c' = \begin{cases} c, & j = 0, \\ 1 - c, & j \neq 0, \end{cases} \qquad A|x\rangle = \frac{1}{\sqrt{60}} \sum_{e,c} |x, e, c\rangle.$$

Every arc is checked: $S^2 = I$, with 300 exchanged pairs and 3360 fixed arcs. The outgoing native record (y, e, c, j) is recovered bijectively from the reversed arc. Thus $A^\dagger A = I$ and $A^\dagger S A = K$, where K is the actual symmetric native transition kernel. With $\Pi = AA^\dagger$ and $W = S(2\Pi - I)$, these are the hypotheses of the abstract quantum-walk spectral mapping theorem: $d_A = A^\dagger$, its discriminant is K, and its unitary is W [111]. Szegedy's corresponding product of two reflections is W^2, whose eigenphase has twice the value [148].

The standard identity $A^\dagger W^n A = T_n(K)$, with T_n the Chebyshev polynomial, attaches to the centered native loads $X_{x,p} = x_p - 1/6$:

$$KX = X(I - L/60), \qquad U_{n+1} - 2U_n + U_{n-1} = -U_n L/30, \qquad U_n = A^\dagger W^n A X.$$

These are compressed amplitudes, not probabilities or measured mean loads. The parameter n counts the supplied coherent continuation, with no identification as physical duration. The binary domain is essential: $(2, 0, \ldots) \mapsto (1, 1, \ldots)$ on the full 78-state carrier has no native reverse event. Its asymmetric kernel cannot equal $A^\dagger S A$ for an involutive unitary. This excludes that same-domain construction, not a larger history-space dilation.

Two exact controls isolate the selection problem. For $|z| = 1$, the unitary $W_z = S[\Pi + z(I - \Pi)]$ has the same first coherent event $W_z A = SA$, but

$$A^\dagger W_z^2 A = K^2 + z(I - K^2).$$

For a fixed basis input, a complete added Ramsey read gives two-step plus probability 1 for $z = 1$ and $113/150$ for $z = -1$, retaining both quantum outputs. Separately, multiplying each coherent record branch on seam $(0, 1)$ with current $+1$ by -1 preserves every committed C and E CP branch, including on arbitrary references. Its twisted embedding A_η nevertheless gives $\mathrm{Tr}[(A_\eta^\dagger S A_\eta)^3] - \mathrm{Tr}(K^3) = -1/900$. Both real continuations conserve load, respect all sixty proper carrier rotations, fix the uniform coherent state and admit the same algebraic time reversal $\Theta = S$ conjugation. Their coins preserve $\|(I - \Pi)v\|$. Under the additional hypothesis of full event-permutation covariance, real orthogonal coins fixing the uniform vector are identity or reflection. A separate sign change on every moving arc preserves proper covariance and the committed instruments, but its discriminant $D = 2\,\mathrm{diag}(K) - K$ satisfies $\mathrm{Tr}(D^3) - \mathrm{Tr}(K^3) = -1/90$ and does not fix the uniform coherent ket. These are restrictions on selection, not proofs of A1–A3 source admission.

Scalar record phases cancel throughout finite adaptive committed-record histories, including arbitrary references. A source-executed interface test instead reads the low unread record bit

in the fixed X basis and the others in Z [25]. For the same preparation, its complete X-plus probabilities are 3/4 and 1/4; committing first gives 1/2 for both. Both outcomes of the fair coin and every output are retained. The coherent protocol does not retain a native-current record alongside the complementary read. The identity $K'_{c,r} = (-1)^{r \bmod 2} K_{c,r}$ holds on the full 512-dimensional encoded carrier, including padding and arbitrary references; executed matrix residuals are numerical diagnostics. Each attempt uses 5380 C or 5383 E source pulses. The two phase-service pulses have zero angles in the control; the summed absolute source-pulse angles differ by π. Coherent access and its implementation are supplied interface obligations; the test selects no phase and supplies no physical calibration or data.

4.7 The local event clock

N4 is the name, in the construction's candidate register, of a reciprocal extension of the binary repair law [15]. Each port carries a private six-level response factor. One coherent exchange $W_e = \exp[\theta(Y \otimes \Gamma_e + X \otimes \Delta_e)]$, with $X = S_e(N_a - N_b)^2$, $Y = iS_e(N_a - N_b)$ and response generators Γ_e, Δ_e built from the existing endpoint response matrices, conjugates the binary repair, $K_r = W_e(L_r \otimes I)W_e^\dagger$. The coupling θ is supplied. Private preparations then change bare load probabilities; at fixed bare load and $\theta = 1/8$ an endpoint occupation probability moves from 0.482020673 to 0.517979327. An interval certificate of the existing N4 construction, produced by the native-response certificate of the repository, proves an edge-Gram margin $\mathcal{G} \geq mI$ at $\theta = 1/16$ and $\theta = 1/8$ [15, 40, 1].

Fix an observer port v. A tick is any attempted seam incident to v, including attempts that leave its load unchanged. Under independent uniform seam selection a tick has probability $p = 1/6$ per attempt, and remote attempts continue between ticks. In independent trials the full tracial state is prepared, the binary occupation N_v is measured, the unmodified source runs until one or two ticks have occurred, and N_v is measured again, with no intermediate measurement, reset or postselection. With $f = N_v - 1/6$ and $\|f\|_\tau^2 = 5/36$, write $c_k = \mathbb{E}[(X_0 - \frac{1}{6})(X_k - \frac{1}{6})]/(5/36)$ for the endpoint correlation after k ticks [40, 41]. The two numbers of Theorem 4.7(iii) are certified lower bounds, the full N4 two-tick correlation is not evaluated, and preparation, incident-event detection and bare-occupation readout are supplied operations that the current detector of Section 4.1 does not implement.

Theorem 4.7 (Memory at observer-local ticks)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be the averages over the thirty seams of the remote and incident seam channels, so that the first-hit channel is $\mathcal{K} = \mathcal{B}(I - \mathcal{A})^{-1}$. Then:*

(i) $c_k = \langle f, (\mathcal{K}^*)^k f\rangle / \|f\|^2$, *and* $c_1 = 1 - 6\alpha_{\mathrm{occ}}/11$ *for the occupation compression factor* α_{occ}*;*

(ii) the statistic

$$b_{\mathrm{tick}} = c_2 - c_1^2 - \frac{r^2(209r + 180)}{1331r^2 + 3300r + 1800}, \qquad r = 1 - c_1,$$

vanishes for the native law and for the one-term exchange controls, which have $c_1 = 5/11$ *and* $c_2 = 93/407$*;*

(iii) for both N4 completions, built on C *and on* E, $b_{\mathrm{tick}} \geq 3m/50$, *and the certified lower bounds are*

$$b_{\mathrm{tick}} \geq \frac{139996257}{12500000000000} \approx 0.00001119970056 \quad (\theta = 1/16),$$
$$b_{\mathrm{tick}} \geq \frac{11335290321}{50000000000000} \approx 0.00022670580642 \quad (\theta = 1/8);$$

(iv) $c_{2,\mathsf{C}} \geq c_{2,\mathsf{E}}$ *and* $b_{\mathrm{tick},\mathsf{C}} \geq b_{\mathrm{tick},\mathsf{E}}$.

Proof sketch. Each seam channel is a positive semidefinite contraction of Hilbert-Schmidt space, with spectrum in $\{0, 1/2, 1\}$ for C and $\{0, 1\}$ for E; N4 conjugation preserves this. Hence $0 \le \mathcal{A} \le (1-p)I$, the first-hit series converges, and the stopped maps are unital and trace preserving. Remote seams fix f, which gives (i). The closed comparator with seam mean maps $I - \frac{\alpha_{\mathrm{occ}}}{2} w_e w_e^{\top}$ yields the correction term in (ii) after substituting $\alpha_{\mathrm{occ}} = 11(1-c_1)/6$. With the Schur complement S of $I-\mathcal{A}$ on the centered occupations, $S \le I$ and $b_{\mathrm{tick}} = \langle z, S^{-1} z\rangle/(p\|f\|^2) \ge \|z\|^2/(p\|f\|^2)$. The hidden defect is $z = -\frac{1}{60}\sum_e t_e S_e$ with five incident coefficients ± 1, so $\|z\|^2 \ge 5m/3600$ and $b_{\mathrm{tick}} \ge 3m/50$. On each seam $\mathsf{C} - \mathsf{E}$ is positive semidefinite, which gives (iv) [40]. □

The correction used for global attempts, $(1 - c_1)^2/10$, assigns $-16/2035$ to the native null at local ticks; the correction of Theorem 4.7(ii) is the one adapted to this stopping instrument. The receipt and eight independent checks bind the result, including a comparison with the full 66-state native stopping kernel and a complete dimension-six quantum model of the resolvent and Schur identities [40, 66].

5 Fields

5.1 The scalar detector and its refinement transfer

The scalar detector joins the native observer to the free-scalar continuum family of the theorem library [30, 31]. The scalar action, quantization, vacuum, preparation, Born readout and model clock are supplied, and native integer averaging does not derive them. The field is the parent's massive scalar on the golden tensor grids $q \in \{13, 34, 89, 233\}$, with Dirichlet boundary and mass one. After mass whitening its one-particle operator is $\mathsf{A} = I + B \otimes I \otimes I + I \otimes B \otimes I + I \otimes I \otimes B$, with the parent's one-dimensional stiffness B; then $\Omega = \sqrt{\mathsf{A}}$ and $H = d\Gamma(\Omega)$ with $\hbar = 1$. A preparation profile f in a compact left slab and a detector profile g in a compact right slab are inherited, both including the square roots of the product masses.

One trial runs as follows. (1) Prepare one of the 66 binary configurations uniformly, repair $(0, 1)$ and retain (c_1, j). (2) Prepare the field vacuum and apply the momentum displacement of amplitude $\gamma j f$ with $\gamma = 4$. (3) Evolve to the model time t and read a centered binary Weyl instrument with

$$K_y = \tfrac{1}{2}[e^{-iX/2} - iy\, e^{iX/2}], \qquad X = a^{\dagger}(v) + a(v), \qquad v = \Omega^{-1/2} g/\sqrt{2},$$

so that $K_y^{\dagger} K_y = \frac{1}{2}[I + y \sin X]$. (4) Commit y, repair $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, and halt. The window of model times is $[19/20, 1]$, with primary setting $39/40$; model time is ct/L for the declared box and carries no laboratory calibration.

Proposition 5.1 (Scalar feedback and refinement transfer)**.** *For the process above at model time t, let $r(t) = \Pr(y = +1 \mid j = 1) - \frac{1}{2}$. Then:*

(i) the branch operators $M_{(r,y,s)} = (L_s L_r) \otimes (K_y e^{-iHt} V_j)$, with V_j the controlled displacement, form one CPTP instrument on arbitrary joint source and field inputs, and each branch retains its conditional field state as a superposition of two coherent components with their relative phase;

(ii) $p(y \mid j) = \frac{1}{2} + yj\, r(t)$ with $r(t) = \frac{1}{2} e^{-\|v\|^2/2} \sin \mu(t)$ and $\mu(t) = \gamma \langle g, \Omega^{-1} \sin(\Omega t) f\rangle$;

(iii) over the 528 histories, $\mathbb{E}[j_2] = 5r/132$, $\mathbb{E}[N_{0,\mathrm{final}}] = \frac{1}{6} - 5r/132$, $\mathbb{E}[yj] = 10r/33$ and $\Pr(y = \pm 1) = \frac{1}{2}$; disconnecting the preparation or replacing y by an independent fair bit gives $\mathbb{E}[j_2] = 0$ with the same seam frequencies, and reversing the kick reverses the sign;

(iv) if $|r_q - r_\infty| \le \varepsilon_q$ *on the window, then the complete labelled histories satisfy the total-variation distance* $\mathrm{TV} = \frac{5}{33}|r_q - r_\infty| \le 5\varepsilon_q/33$*, and* $|\Delta\mathbb{E}[j_2]| \le 5\varepsilon_q/132$.

Proof sketch. The Weyl operators $e^{\pm iX/2}$ are displacements by $\pm iv/2$, which gives the two-component branch states and (ii); the zero-current branch has probability one half for either record. Conditioned on the initial configuration and first coin, the two detector laws differ in total variation by $|j|\,|r_q - r_\infty|$, and the subsequent native update is common; the expectation bounds use the exact affine formulas [31]. □

At $q = 233$ the parent certifies, uniformly for $19/20 \le t \le 1$,

$$\varepsilon \le \frac{12513633763}{250000000000}, \qquad -\frac{111322580247}{500000000000} \le r \le -\frac{34438518047}{250000000000}.$$

The finite-grid mean second current therefore satisfies

$$-\frac{37107526749}{4400000000000} \le \mathbb{E}[j_2] \le -\frac{34438518047}{6600000000000},$$

negative throughout the window, and the continuum mean lies within $\frac{5}{132}\varepsilon \approx 0.001896$ of it [67, 32]. At the primary time 39/40 the computed plus probabilities for $j = +1$ are approximately 0.334515, 0.319421, 0.317093 and 0.316771 on $q = 13, 34, 89, 233$, and the finest-grid prediction is $\mathbb{E}[j_2] \approx -0.00694050$. A seeded execution with 200,000 fresh trials per grid (seed 20261012) tests Born outcomes against predictions computed before sampling, with paired reversal, independent-donor and fixed-seam controls. At $q = 233$ all 12,487,168 field modes contribute; no mode projection or occupation cutoff replaces the field. The two coarse grids carry the parent's trivial bound $\varepsilon = 1$. An independent check diagonalizes the family's weighted Dirichlet operator on its unit cube. For four fixed standing-wave labels the largest relative frequency error against the supplied massive dispersion law $\omega^2 = 1 + \pi^2(n_x^2 + n_y^2 + n_z^2)$ is 2.137%, 0.2875% and 0.04116% at $q = 13$, 34 and 89 [59]. The labels are standing waves on a box, and the frequencies follow from the supplied quadratic action.

A local extension replaces continuous propagation by canonical kick-drift-kick updates $\mathbf{p} \leftarrow \mathbf{p} - \frac{\tau}{2}\mathsf{A}\mathbf{u}$, $\mathbf{u} \leftarrow \mathbf{u} + \tau\mathbf{p}$, $\mathbf{p} \leftarrow \mathbf{p} - \frac{\tau}{2}\mathsf{A}\mathbf{u}$ of the field configuration $\mathbf{u}$ and momentum $\mathbf{p}$, where A acts through the six-neighbour stencil [31]. It runs at $q = 13$ and $q = 34$ at the certified update nearest 39/40 and retains the attained rational time. The detector variance $\frac{1}{2}[a_n^\top \Omega^{-1} a_n + b_n^\top \Omega b_n]$, computed from the propagated pullback columns and the original vacuum, includes the squeezing caused by split evolution, and a $q = 13$ step-halving control compares at the same attained time. Replacing ε_q by the parent's combined spatial and temporal bound transfers through the same factors 5/33 and 5/132. At $q = 233$ the certified temporal term is at most 0.000008071512 and the combined grid-time probability error at most 0.050062606564, both analytic bounds for the ideal split dynamics [30].

5.2 The retained two-read continuation

The continued observer keeps the disturbed field after its first read [33]. One source and one field are prepared per history. The field is read at model times 19/20 and 1; the first bit y_1 selects the second native seam, the conditional field evolves to time 1, the second bit y_2 selects the third seam, and the observer halts with five records readable. This continuation uses exact continuous spectral evolution, so the local split-update engine of Section 5.1 and this two-read history carry separate error bounds.

Proposition 5.2 (Two reads of one field)**.** *In the continued protocol:*

(i) all 2,112 labelled histories are retained; the record-prefix counts at the six phases are $1, 4, 8, 24, 48, 124$*, so the 209 reached prefixes have an eight-bit encoding and the 124 terminal words need seven bits;*

(ii) with the Gram kernel G_{ab} *of the two detector vectors,* $V = G_{11} = G_{22}$*,* $C = \operatorname{Re} G_{12}$*,* $c = 2 \operatorname{Im} G_{12}$ *(so that* $[X_1, X_2] = ic$*),* $S_a(j) = e^{-V/2} \sin(j\mu_a)$*,* $D = \cos(c/2)$ *and* $R_j = \frac{1}{2}[e^{-(V-C)} \cos(j(\mu_1 - \mu_2)) - e^{-(V+C)} \cos(j(\mu_1 + \mu_2))]$*, the joint law is*

$$p_j(y_1, y_2) = \tfrac{1}{4}[1 + y_1 S_1(j) + y_2 D\, S_2(j) + y_1 y_2 R_j];$$

(iii) on the positive-current preparation, performing the first read changes the second plus probability by $(-r(t_2))[1 - \cos(c/2)]$*, certified greater than* 0.000022281162 *at* $q = 233$ *and greater than* 0.0000252366 *in the declared compact continuum; a commuting Gaussian control with the same means and real covariance has* $D = 1$ *and no such disturbance; the zero-current vacuum correlation exceeds* 0.045619388905 *at* $q = 233$*;*

(iv) the total variation distance of the complete labelled history from the declared continuum is at most 1*,* 0.451806*,* 0.052189606637 *and* 0.025709913643 *on* $q = 13, 34, 89, 233$*, including the zero-current histories.*

Proof sketch. A state is stored as $\sum_n c_n D(i \sum_a h_{na} v_a)|\alpha\rangle$ with exact half-integer shifts h; its Gram matrix is explicit in G and the mean vector, and every one of the field modes enters G. Both terms of each Kraus operator are applied before the branch norm is taken. The closed law follows from the coherent characteristic function. For four atoms with signed Walsh-moment differences a, b, d, $\mathrm{TV} = \frac{1}{2} \max(|a|, |b|, |d|, \frac{1}{2}(|a| + |b| + |d|))$; the parent's one-particle residuals and compact tails bound the moment differences in directed rational arithmetic [33]. □

Numerical execution at $q = 233$ gives $\mathbb{E}[j_3] \approx -0.00130126$ and $\mathbb{E}[y_1 y_2 \mid j = 0] \approx 0.0514838$; deleting the first field record from the forecast raises the model's Brier risk by about 0.00064677. The software run uses 50,000 trials per grid with seed 20261012 and does not resolve the small disturbance statistically. The conditional-field donor, the field reset, the commuting control, the independent bit and the disconnected actuator separate correlation from measurement disturbance.

5.3 The source-compiled two-site apparatus

The two-site apparatus is a separate finite field design whose preparation, evolution and pointer all execute emitted twelve-port control words [34, 35]. Its action lies outside the golden refinement family, so its occupation limit carries no spatial refinement. The Hamiltonian is

$$H = \tfrac{1}{2}(P_0^2 + P_1^2 + Q_0^2 + Q_1^2) + \tfrac{\kappa}{2}(Q_0 - Q_1)^2, \qquad \kappa = \tfrac{9}{32},$$

with normal-mode frequencies 1 and 5/4. Each site uses $d \in \{4, 8, 16\}$ number states. The executed Hamiltonian uses the compressed physical squares $Q_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$ and $P_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$, so that $H_d = \Pi_d H \Pi_d$; squaring truncated quadratures would implement a different Hamiltonian, and $[Q_d, P_d] = i(I - d|d-1\rangle\langle d-1|)$ forbids treating the finite gates as exact Gaussian shears.

The prepared state is the normalized projection of the coupled vacuum onto total occupation at most $d-2$, compiled from blank registers into a source word. The eight-qubit register of $d = 16$ uses a binary amplitude tree with commuting Walsh-Y factors, which costs $3((n-1)2^n + 1)$ native

pulses for n qubits, 5,379 at $n = 8$. The current impulse is $e^{ijQ_{0,d}}$; 64 or 128 symmetric split steps reach the model coordinate $t = 1$; the pointer is prepared in $|+\rangle$, coupled by $e^{-iZQ_{1,d}/2}$ and read in the Y basis, with field effects $\frac{1}{2}(I \pm \sin Q_{1,d})$. Only the pointer bit reaches the controller. Parity gives $\Pr(y = +1 \mid j) = \frac{1}{2} + jr_d$, so $\mathbb{E}[j_2] = 5r_d/132$ as in Proposition 5.1.

Theorem 5.3 (Complete ideal error budget of the two-site instrument)**.** *For the declared two-site action, model time $0 \le t \le 1$ and native current $j \in \{-1, 0, 1\}$:*

(i) with A the onsite and B the bond part of H_d and $S(h) = e^{-ihA/2}e^{-ihB}e^{-ihA/2}$,

$$\|S(t/n)^n - e^{-it(A+B)}\| \le \frac{t^3}{n^2}\Big(\frac{\|[A,[A,B]]\|}{24} + \frac{\|[B,[A,B]]\|}{12}\Big),$$

uniformly over inputs and references;

(ii) the prepared state differs from the infinite coupled vacuum in trace distance by less than 0.007576, 0.000079899 and 1.0324×10^{-8} at $d = 4, 8, 16$;

(iii) the trace distance between the complete ideal finite instrument, including its classical outcome and unnormalized terminal field branches, and the infinite two-oscillator instrument is below 0.799447, 0.077171 and 0.000196 at $d = 4, 8, 16$, and the bound tends to zero as d grows;

(iv) at $d = 16$ and $n = 128$ the combined occupation and splitting error is below 0.001464436, and the ideal source-word instrument satisfies $0.514957066204 \le \Pr(y = +1 \mid j = 1) \le 0.517885937293$, an interval that excludes $1/2$.

Proof sketch. Item (i) is a direct defect bound for the symmetric product formula [89], with nested-commutator norms bounded by exact rational Frobenius norms in the factorial-weighted number basis. For (ii), the vacuum overlap is $\sqrt{c_J}$ with an explicit squeezed tail. For (iii), the infinite state is a displaced squeezed vacuum throughout; the Gaussian generating function $\langle z^N \rangle$ bounds the number tails uniformly along impulse, evolution and both pointer arms; the boundary residual of the projected generators is controlled by those tails; projected Duhamel comparison and the orthogonal terminal leakage give $e_{\text{full}} \le (e_{\text{proj}}^2 + T_1)^{1/2}$ without dividing by any Born probability. Item (iv) combines (i), (iii) and a rational enclosure of the infinite reference probability obtained from alternating sine series and a Taylor bound on $e^{9/40}$ [35, 68]. □

Table 6 lists the executed plus probabilities for a positive unit current with their splitting bounds. The independent continuous finite Hamiltonian gives approximately 0.516422345746 at $d = 8$ and 0.516421501748353 at $d = 16$. The infinite two-oscillator reference has detector variance $9/20$ and mean $\mu(t) = \frac{1}{2}(\sin t - \frac{4}{5}\sin(5t/4))$ for unit impulse, so $\Pr(y = +1 \mid j) = \frac{1}{2}(1 + e^{-9/40}\sin(j\mu(t)))$, equal to 0.516421501748 at $t = 1$ and $j = 1$ [34].

The $d = 16$, 128-step protocol uses nine work qubits, 18,612,125 native pulses and 37,224,250 proper-code load and export events; the executor contracts each generated subroutine once, so these are logical operation counts. The complete export channel includes all failure outcomes, which have zero probability on the invariant valid-code histories in exact arithmetic. Retained 200,000-trial native runs record the detector response, terminal branches, complete record counts and the reversal, disconnected and donor controls [34].

Table 6: Two-site apparatus: source-executed plus probability for a positive unit current and the exact rational finite splitting bound, rounded upward [34].

Onsite levels d	Steps	Executed $\Pr(y=+1)$	Splitting bound
4	64	0.516680958	0.000103957
4	128	0.516679563	0.000025990
8	64	0.516424174	0.000693013
8	128	0.516422803	0.000173254
16	64	0.516423330	0.005076875
16	128	0.516421959	0.001269219

6 The common interacting source

6.1 Construction

The common source places quantum electric flux, a charged fermion pair and a neutral scalar on one evolving work register. The original sharp protocol measures it twice and feeds its records into the register and the declared native integer primitive [42, 43]. Its four-read smooth extension joins compiled native quantum channels and encoded recovery (Section 6.6). The action, preparation, coupling and model-coordinate schedule remain construction inputs.

Order four fermion modes as L_0, R_0, L_1, R_1. The two channels carry charges $q_0 = +1$ and $q_1 = -1$ and one fermion each; $b_j = 0$ means left and $b_j = 1$ right occupation of channel j. The electric link has basis $|\ell\rangle$, $\ell \in \mathbb{Z}$, with $\mathcal{E}|\ell\rangle = \ell|\ell\rangle$ and $U|\ell\rangle = |\ell+1\rangle$. The isometry

$$W|b_0, b_1\rangle = |\text{modes } b_0 \text{ and } 2+b_1 \text{ occupied}\rangle \otimes |-b_0+b_1\rangle$$

maps onto four orthonormal occupation and flux vectors.

Proposition 6.1 (Flux-dressed matter code)**.** *On the image of W:*

(i) both operator Gauss constraints hold, $G_L W = G_R W = 0$ with $G_L = \mathcal{E} - Q_L$ and $G_R = -\mathcal{E} - Q_R$, the supported flux set is $\{-1, 0, 1\}$ inside the infinite link, and fermion parity is even;

(ii) the dressed hopping terms $h_j = c_{Lj}^\dagger U^{q_j} c_{Rj} + c_{Rj}^\dagger U^{-q_j} c_{Lj}$ preserve both ambient Gauss values and the code, with $W^*(h_0 + h_1)W = X_0 + X_1 = H_{\mathrm{hop}}$ and $W^*\mathcal{E}^2 W = \frac{1}{2}(I - Z_0 Z_1)$, and $\|[H_{\mathrm{hop}}, \mathcal{E}^2]\| = 2$;

(iii) with $\theta = \operatorname{atan2}(4,3)$, $u = (3I - 4iX)/5$, $U_\theta = u \otimes u = e^{-i\theta H_{\mathrm{hop}}}$ and $V_\theta = e^{-i\theta\mathcal{E}^2} = \operatorname{diag}(1, z, z, 1)$, $z = (3-4i)/5$, the echo $U_\theta^* V_\theta U_\theta$ applied to $|00\rangle$ gives right occupation of channel 0 with probability exactly $576/3125$, and without V_θ the probability is zero.

Proof sketch. On every image column $Q_L = -b_0 + b_1 = \mathcal{E}$ and $Q_R = -\mathcal{E}$. The full four-mode algebra of canonical anticommutation relations (CAR) is checked on all 16 occupation states, with independent Jordan-Wigner signs; right-to-left motion raises the flux by q_j and left-to-right motion lowers it. All entries of the echo unitaries are Gaussian rationals, and $U_\theta^* V_\theta U_\theta |00\rangle = \frac{1}{3125}(2549 - 1152i,\ 336 - 168i,\ 336 - 168i,\ -576 - 1152i)$ [43]. □

The echo runs on the source controls with 22 native pulses and 44 coherent service events. Two neutral scalar sites join the same register. With the onsite quadratics compressed from the infinite

oscillator, the declared dimensionless action is

$$H = H_{\rm hop} + \mathcal{E}^2 + H_{\rm scalar} + H_{\rm int},$$
$$H_{\rm scalar} = \sum_{j=0,1} \tfrac{1}{2}[P_j^2 + (1+\kappa)Q_j^2] - \kappa Q_0 Q_1, \qquad H_{\rm int} = g\sqrt{2}\, Q_0(n_{L0} - n_{R0}), \tag{3}$$

with $\kappa = 9/32$ and $g = 1/4$. The interaction commutes with both Gauss generators and with fermion parity, and it couples the sectors: $[H_{\rm int}, H_{\rm hop}] = 2ig\sqrt{2}\, Q_0 Y_0$. At $d = 2$ the whole action has rational entries,

$$H = X_0 + X_1 + \tfrac{1}{2}(I - Z_0 Z_1) + \tfrac{73}{64}(n_{s0} + n_{s1} + 1) - \tfrac{9}{64} X_{s0} X_{s1} + \tfrac{1}{4} X_{s0} Z_0,$$

where the index s marks the scalar sites.

The experiment proceeds in four steps. (1) Native repair on $(0,1)$ records a current j, which controls the emitted gauge pulse $V_j = e^{-ij\theta X_0}$. (2) The register evolves for model-coordinate interval 1 (U_1); a source pointer measures the right occupation of the first charged species and commits a bit a (P_a). (3) Only a selects the hopping pulse $F_1 = (3I - 4iX_0)/5$, with $F_0 = I$; the register evolves for interval $1/2$ (U_2). (4) A second pointer measures the sign of Q_0 and commits b (R_b); only b selects the second native seam, $(0,5)$ for $b = 1$ and $(0,1)$ otherwise.

Proposition 6.2 (One joint instrument with retained histories)**.** *The operators* $K_{(a,b)} = R_b U_2 F_a P_a U_1$ *satisfy* $\sum_{a,b} K_{(a,b)}^\dagger K_{(a,b)} = I$, *and* $M_{(r,a,b,s)} = (L_s^{e(b)} L_r^{(0,1)}) \otimes (K_{(a,b)} V_j)$ *form one CPTP instrument on arbitrary joint native and quantum inputs. Under the uniform binary preparation and the declared quantum vector* ψ, *the* $66 \cdot 2 \cdot 4 \cdot 2 = 1{,}056$ *histories have probabilities* $p(a,b \mid j)/264$ *with* $p(a,b \mid j) = \|K_{(a,b)} V_j \psi\|^2$, *each bound to its terminal density* $|v_{j,a,b}\rangle\langle v_{j,a,b}|/264$, *and the first native record keeps the masses* 14/33, 14/33, 5/66, 5/66. *For a complete record-controlled instrument* $B_y = \sum_r |r\rangle\langle r| \otimes K_{(r,y)}$ *with channel* $\Lambda(X) = \sum_y B_y X B_y^\dagger$, *every old record projector* Q_r *satisfies* $\Lambda^*(Q_r) = Q_r$, *and the block readback* $E_{\rm block}(X) = \sum_r Q_r X Q_r$ *and the public readback* $E_{\rm public}(X) = \sum_r \mathrm{Tr}(Q_r X Q_r) Q_r / \dim W$ *are distinct conditional expectations.*

Proof sketch. Summing $M^\dagger M$ over s, then (a,b), then r gives the identity. Each B_y commutes with Q_r, so $\Lambda^*(Q_r) = Q_r \sum_y B_y^\dagger B_y = Q_r$, the dual fixed-observable criterion of the record theorems; the joining code checks every reported probability against the norm of the retained unnormalized vector [43, 69]. □

At 512 steps per interval the emitted words use 4 working qubits, 176,182 native pulses and 352,364 coherent load and export events at $d = 2$, and 6 qubits, 1,879,628 pulses and 3,759,256 events at $d = 4$. Disconnected and feedback-disabled controls, a fixed-seam control and an independent actuator bit with the same unconditional frequency isolate the causal effect of the records [42].

6.2 The finite feedback contrast

Theorem 6.3 (Finite feedback contrast)**.** *For the common source with intervals* 1 *and* 1/2*:*

(i) every palindromic product $F(h)$ *of the component exponentials whose generators sum to* H *satisfies* $\|F(t/n)^n - e^{-itH}\| \le S^3 |t|^3/(3n^2)$ *with* $S \le 3 + (1+2\kappa)(2d-1) + |g|\sqrt{4d-2}$, *on all inputs and references; at* $n = 512$ *and* $t = 1$ *the half-diamond bounds are below* 0.000728 *for* $d = 2$ *and* 0.004184 *for* $d = 4$*;*

(ii) at $d = 2$ and zero native preparation current, exact rational Taylor arithmetic of degree 64 with $\|H\| \le 109/16$ encloses the unconditional probability of a positive scalar coordinate and the feedback contrast for the continuous finite Hamiltonian:

$$\begin{aligned} &\textit{feedback enabled:} && [0.370697726608684,\ 0.370697726608685],\\ &\textit{feedback disabled:} && [0.402359206973491,\ 0.402359206973492],\\ &\textit{contrast:} && [-0.031661480364808,\ -0.031661480364807]; \end{aligned}$$

(iii) for the ideal source word at 512 steps per interval the contrast lies in

$$[-0.033297300667839,\ -0.030025660061775],$$

which excludes zero.

Proof sketch. For (i), the multinomial Leibniz expansion bounds the third derivative of a product of unitaries by S^3, palindromic symmetry makes the product agree with e^{-ihH} through second order, and unitary telescoping gives the displayed bound. For (ii), the integral Taylor remainder contains a unitary, so $\|T_m(t) - e^{-itH}\| \le (St)^{m+1}/(m+1)!$; branch vectors carry error at most $\eta = \epsilon_2 + (1+\epsilon_2)\epsilon_1$, each probability error is at most $4\eta + 2\eta^2$ and the contrast error at most $8\eta + 4\eta^2$; at $m = 64$ the probability radius is below 7.091×10^{-37}. Inserted measurements and feedback contract trace distance, which gives (iii) [43]. □

With $g = 0$ the gauge-only feedback cannot change the disconnected scalar marginal, and that control runs with the same requests and counted services. A coarse splitting or an inadequate Taylor degree yields an interval that contains zero, and the evaluator then reports no resolved signal.

6.3 The infinite-occupation contrast

The finite pointer measures $B_d = \mathbf{1}_{(Q_d>0)}$, the sign projector of the truncated coordinate, whereas the infinite detector multiplies by $\mathbf{1}_{x_0>0}$. Compression and spectral sign do not commute. At $d = 2$, $\mathbf{1}_{(Q_2>0)} = \frac{1}{2}(I + X)$ while $V_2^* \mathbf{1}_{(Q>0)} V_2 = \frac{1}{2} I + X/\sqrt{2\pi}$, and the two differ in norm by $\frac{1}{2} - 1/\sqrt{2\pi} \approx 0.10105772$ [47]. The finite signal of Theorem 6.3 is a statement about its own detector; the infinite-occupation statement needs the infinite detector.

Theorem 6.4 (Infinite-occupation feedback contrast)**.** *Take the action (3) on two infinite oscillator Fock spaces, the infinite free coupled vacuum, proportional to* $\exp[-(a_0^\dagger - a_1^\dagger)^2/36]|0,0\rangle$ *with squared normalization* $4\sqrt{5}/9$*, the blank gauge code, zero native preparation current, the matter record and feedback of Section 6.1, and the half-line readout of* Q_0*. Then:*

(i) the unconditional positive-coordinate probability with feedback minus that without lies in $[-0.024045670143,\ -0.024043450955]$*, and without the interaction the contrast is exactly zero;*

(ii) both matter records and both half-line outcomes retain their quantum states, including the post-read infinite occupation tail;

(iii) along $d = 2^k$, with exact palindromic factors and $n = d^2$ steps per interval, the complete finite sign instruments converge in trace distance to this infinite instrument, terminal states included, and the finite-family contrast is strictly negative for all sufficiently large d.

Proof sketch. The linear force is relatively bounded by the positive coupled oscillator Hamiltonian K, so H is self-adjoint on $\mathcal{D}(K)$, and finite Hermite polynomials lie in every $\mathcal{D}(H^k)$. Each of 24 steps of length $1/16$ applies a degree-16 Taylor polynomial of the full H on expanding support, charges its remainder $\delta^{17}\|H^{17}v\|/17!$ exactly, then discards support above total occupation 16 and rounds coefficients to a 2^{-112} grid, charging both differences; the largest expanded occupation is 50. The complete output then has half trace-norm error at most $\tau + \epsilon + \epsilon^2/2$ and each event at most $\tau + 2\epsilon + \epsilon^2$. Half-line probabilities have the form $r + s/\sqrt{2\pi}$ with exact rationals r, s, and π is enclosed by Machin's formula. For (iii), $A_d = P_d Q_0 P_d$ converges to Q_0 in the strong resolvent sense on the Hermite core, the spectral measure of Q_0 has no atom at zero, and $\det Q_{2m} \neq 0$, so the sign projectors converge strongly [44, 90, 87]. □

The $d = 2$ finite continuous experiment has contrast about -0.03166148; replacing only its detector by the infinite half-line effect gives about -0.02526221; removing the dynamical cutoff as well gives the interval of Theorem 6.4 [44]. The signal survives both limits, and both approximations change its size. The interval is specific to zero native preparation current and the declared two-site action.

6.4 Occupation refinement and complete-output comparisons

Let P_d project both oscillators onto occupations below d and let $N = n_0 + n_1$ be the total oscillator occupation, a different object from the closure input N. Two moment estimates control the history before the final read. The first is uniform over matter inputs.

Theorem 6.5 (Uniform pre-readout occupation bound)**.** *Let the matter start in an arbitrary state entangled with an untouched reference, let the scalar start in the fixed coupled vacuum, and let the finite preparation retain total occupations* $0, 2, \dots, d-2$*. Then:*

(i) $M(s,t) = \langle e^{sN} \rangle$ *satisfies* $\partial_t M \leq \frac{11}{16} \sinh(s)\, \partial_s M + \frac{25}{32} \sinh(s)\, M$ *through the matter measurement and feedback;*

(ii) at total model time $T = 3/2$ *the resulting moment bound holds with base* $\varrho = 113/64 \geq 7/4$ *and* $M < 13$*, where* $\varrho \leq e^{s(T)}$*;*

(iii) $\|P_d H (1 - P_d)\| \leq C_d = \frac{\kappa}{2}\sqrt{d(d+1)} + \frac{3\kappa}{2} d + \frac{1}{4}\sqrt{d}$*, and the retained-history trace distance up to the final scalar read is at most* $\left[(\sqrt{2v_d} + T C_d \sqrt{\tau_d})^2 + \tau_d\right]^{1/2}$ *with* $\tau_d = M/\varrho^d$ *and* v_d *the omitted preparation probability, an* $O(d\, \varrho^{-d/2})$ *rate;*

(iv) this bound is below 0.011408*,* 0.000002498 *and* 6.175×10^{-14} *at* $d = 32, 64, 128$*.*

Proof sketch. A passive rotation to common and difference modes leaves only the squeezing term $\frac{\kappa}{2}(a_-^2 + a_-^{\dagger 2})$ and the force as number-changing terms; weighted commutator and Cauchy inequalities that hold for matter-entangled vectors, followed by Young's inequality, give (i). Along $\tanh(s(t)/2) = e^{-11t/16} \tanh(s_0/2)$ the moment grows by at most $e^{(25/22)(s_0 - s(t))}$ from its initial value $\sqrt{80/(81 - \varrho_0^2)}$, equal to $\sqrt{80/17}$ at $\varrho_0 = 8$. The coherent record dilation and the controlled feedback commute with N. Projected Duhamel comparison and the orthogonal outside tail give (iii) [45]. □

Theorem 6.6 (Fourth moment and strong convergence of the sharp instrument)**.** *Let* $w(N) = (N+2)^4$ *and let* $M(t) = \langle w(N) \rangle$ *be summed over the unnormalized record branches. Then:*

(i) $|\dot{M}| \le CM$ *with* $C = 8\kappa + 4|g| = 13/4$, *and every finite preparation has* $M(0) \le M_0 = 143748821/8192000$;

(ii) the record-dilation vectors immediately before the final read satisfy

$$\|\Psi_d - \Psi\| \le \Big[(e_0 + I_T L_d)^2 + \frac{M_0 e^{CT}}{(d+2)^4}\Big]^{1/2} = O(1/d),$$

with $I_T = \sqrt{M_0}\,(e^{CT/2} - 1)/(C/2)$, $L_d = \kappa/(d+2) + |g|/(d+2)^{3/2}$ *and* e_0 *the preparation error;*

(iii) the sign projectors of $Q_{0,d}$ *converge strongly to those of* Q_0, *and the complete unnormalized classical and quantum history, together with its native continuation, converges in trace norm for all three native currents and both feedback settings.*

Proof sketch. On states with total occupation at most m, $\|H\psi\| \le 4(m+2)\|\psi\|$, and each application of H raises the occupation by at most two, so finite-particle vectors are analytic and H is essentially self-adjoint on them [88]. Core convergence gives strong resolvent convergence and strong convergence of the unitary groups [90]. The coefficients of the moment derivative are bounded by 8κ and $4|g|$, the latter through an explicit positive polynomial certificate; the gauge measurement and the conditional rotations preserve $w(N)$ summed over branches. The spectral measure of multiplication by x has no atom at zero, which gives strong convergence of the sign projectors [46]. □

Strong convergence supplies no finite-d rate for the sharp readout. The complete-output certificate supplies finite-d enclosures at $d \in \{2, 32, 64, 128\}$ for the zero-current preparation [47, 70]; no all-cutoff rate for the sharp readout is proved, and the enclosures are fixed-state statements for that preparation and the named rectangular occupation codes.

Theorem 6.7 (Complete outputs of the sharp detector)**.** *For zero native preparation current, with both matter records and both sign outcomes retained:*

(i) for two complete post-read outputs with rank-one branches, the half trace norm of their difference is $\frac{1}{2}\sum_b \sqrt{(p_b + q_b)^2 - 4|z_b|^2}$, *summed over earlier records, where* p_b, q_b *are the branch weights and* z_b *their overlap;*

(ii) the complete continuous-output distances between the finite and the infinite sharp histories lie in the intervals of Table 7; at $d = 2$ *the ideal 512-step source budget widens them to* $[0.454336, 0.455973]$ *with feedback and* $[0.456824, 0.458461]$ *without;*

(iii) every output state supported in the rectangular occupation code of $d = 256$ *lies at distance above* 0.122 *from the true output of either prepared history;*

(iv) for the vacuum control, whose two sign probabilities are both $1/2$, *the best approximation by a state supported in the code with* $d = 2m$ *has distance between* $\sqrt{3/(8\pi)}\,m^{-1/4}$ *and* $\sqrt{4/(3\pi\sqrt{3})}\,m^{-1/4}$, *hence of order* $d^{-1/4}$, *and distance at most* ε *requires* $d \ge 9/(32\pi^2\varepsilon^4)$.

Proof sketch. Item (i) is the trace norm of a difference of two rank-one operators. For (iii), with inside mass a_{ab} and outside mass t_{ab} of each branch, the operator exchanging their normalized directions is a witness of norm one, so every code-supported output has distance at

least $\sum_{a,b}\sqrt{a_{ab}t_{ab}} - \epsilon_{\mathrm{poly}}$. Half-line matrix elements of Hermite functions are explicit, rational up to a factor $1/\sqrt{2\pi}$ [79]. Larger finite sign projectors are certified through exact eigenbasis residuals, $\|P_A - P_C\|_F \le \delta/\gamma$. For (iv), the omitted vacuum mass per sign branch is $T_d = \frac{1}{4} - \frac{1}{2\pi}\sum_{k=0}^{m-1}\binom{2k}{k}/(4^k(2k+1))$, which obeys $1/(4\pi\sqrt{m}) \le T_{2m} \le 2/(3\pi\sqrt{3m})$ [47]. □

Table 7: Certified complete continuous-output distances between the finite sharp history and the infinite sharp history, zero native preparation current, rounded outward [42, 70].

Onsite dimension d	Feedback enabled	Feedback disabled
2	[0.455154, 0.455155]	[0.457642, 0.457643]
32	[0.203713, 0.226530]	[0.203935, 0.226752]
64	[0.180917, 0.180924]	[0.181090, 0.181097]
128	[0.152201, 0.152204]	[0.152341, 0.152344]

At $d = 64$ the pre-readout error is below 0.000002498, so the sharp detector dominates the final error. Accurate sign probabilities do not certify an accurate post-measurement state; the long occupation tail of the sharp detector forces a large output error even where its probabilities are close.

6.5 The smooth Ramsey instrument

A separate detector variant keeps the preparation, evolution, first record and feedback, and replaces the final sharp read by a source Ramsey pointer [48]. Its work-space Kraus maps are

$$M_0 = e^{-i\pi/4}\cos\left(\tfrac{Q_0}{2} - \tfrac{\pi}{4}\right), \qquad M_1 = e^{i\pi/4}\cos\left(\tfrac{Q_0}{2} + \tfrac{\pi}{4}\right),$$

with effects $\frac{1}{2}(I + \sin Q_0)$ for bit zero and $\frac{1}{2}(I - \sin Q_0)$ for bit one. Bit zero thus denotes the plus effect, a detector meaning different from the sharp sign record, so neither the sharp-sign certificates nor the sharp-history clock carry over to this instrument.

Theorem 6.8 (Smooth instrument)**.** *For the declared initial vacuum family, uniformly over the three native current kicks and both feedback settings, with $T = 3/2$ and the constants of Theorem 6.6:*

(i) the complete recorded instrument, including the pointer interaction and both unnormalized outcomes, lies within trace distance $\min(1, E_d)$ of its infinite counterpart, where $A_T = \sqrt{M_0}e^{CT/2}$ and

$$E_d = \Big[\Big(e_0 + I_T L_d + \frac{A_T(e-1)}{2(d+2)^{3/2}}\Big)^2 + \frac{M_0 e^{CT+2}}{(d+2)^4}\Big]^{1/2} = O(1/d);$$

(ii) with $n(d) = d^2$ symmetric steps per interval the combined ideal temporal and occupation error is $O(1/d)$;

(iii) on normalized inputs in the form domain of the infinite Hamiltonian the unconditional increment of the full energy is exactly $1/8$ in the declared dimensionless convention.

Proof sketch. Along the auxiliary pointer parameter $s \in [0,1]$ the generator $\frac{1}{2}Z_{\mathrm{anc}}Q_0$ obeys $dM/ds \le 2M$, and only its lowering part returns outside occupation to the code; integrating gives the pointer term. For (iii), $f_0(q) = \cos(q/2 - \pi/4)$ and $f_1(q) = \cos(q/2 + \pi/4)$ satisfy $\sum_b f_b^2 = 1$, $\sum_b f_b f_b' = 0$ and $\sum_b (f_b')^2 = \frac{1}{4}$, so the kinetic term gains exactly $\frac{1}{8}\|\psi\|^2$, and every other term of H commutes with $f_b(Q_0)$ [48]. □

The conservative bound is about 0.2063 at $d = 64$, 0.04096 at $d = 256$ and 0.008840 at $d = 1024$; at the executed $d = 2$ and $d = 4$ it is vacuous [42]. The identity of (iii) belongs to the infinite canonical oscillator; at a finite cutoff the commutator defect removes it, while the finite instrument keeps finite energy [48, 59].

6.6 Continued instruments, compiled native repair and encoded memory

One execution engine carries the original sharp and smooth protocols and the following extension, with no intermediate work preparation [49]:

$$U(1),\ P \to r_0,\ F(r_0),\ U(1/2),\ M \to r_1,$$
$$U(1/2),\ M \to r_2,\ F(r_2),\ U(1/2),\ P \to r_3.$$

Here P is the complete gauge instrument, M the smooth instrument of Theorem 6.8, and F the existing gauge feedback. The Hamiltonian model time is 5/2; each smooth pointer separately uses auxiliary parameter one. All sixteen unnormalized work/reference branches remain. Complete Kraus families preserve every old-prefix effect by summing later outcomes; simulator checkpoints bind the input and prefix maps, and imported data are validated by prefix reexecution. This supplies neither physical quantum serialization nor cloning. The classical native attachment enumerates 4,224 joint outcomes at each executed cutoff $d = 2, 4$.

For the declared scalar vacuum and coupling $g = 1/4$, the occupation estimate carries $A \geq \sqrt{\langle (N+2)^4 \rangle}$ and $e \geq \|\Psi_d - \Pi_d \Psi\|$ from one stage to the next. Put $a = 13/8$, $L_H = 9/[32(d+2)] + 1/[4(d+2)^{3/2}]$ and $L_S = 1/[2(d+2)^{3/2}]$. Starting with $A = \sqrt{M_0}$ and the preparation error $e = e_0$, the updates are

$$(e, A) \mapsto \begin{cases} (e + A(e^{at} - 1)L_H/a,\ Ae^{at}), & U(t), \\ (e + A(\mathrm{e} - 1)L_S,\ A\mathrm{e}), & M, \\ (e, A), & P \text{ or } F. \end{cases}$$

Projected Duhamel comparison gives each increment; the complete pointer dilation supplies its moment growth. Only at the final stage is the outside tail added, giving $E_d = [e^2 + A^2/(d+2)^4]^{1/2}$. Thus the complete output distance is at most $\min(1, E_d + \sum_i \delta_i)$, where δ_i is the all-input temporal bound for interval i. At any fixed finite smooth protocol, $n(d) = d^2$ makes this $O(1/d)$. The uniform moment bounds at the executed $d = 2, 4$ remain vacuous. Arbitrary scalar inputs and post-sharp continuations do not inherit this vacuum-family certificate.

A separate state-specific certificate retains all sixteen outputs for the blank-gauge and coupled scalar-vacuum preparation, for each current $j \in \{-1, 0, 1\}$, including both smooth reads, later feedback and the gauge probe [50]. For finite and infinite polynomial branch approximants u_r, v_r, each divided by its initial preparation norm, put $a_r = \|u_r\|^2$, $b_r = \|v_r\|^2$ and $c_r = \langle u_r, v_r \rangle$. The rank-one trace-norm identity and complete-channel contractivity give

$$D_{d,j}^{\text{ideal product},\infty} \leq \frac{1}{2} \sum_{r \in \{0,1\}^4} \sqrt{(a_r + b_r)^2 - 4|c_r|^2} + \epsilon_d + \epsilon_\infty + \sum_i \delta_i,$$

where the ϵ terms bound complete preparation and polynomial-output errors, and the δ_i terms charge finite temporal splitting once. Directed rational evaluation at 512 splits per interval gives

$$\max_{j \in \{-1,0,1\}} D_{d,j}^{\text{ideal product},\infty} < \begin{cases} 0.264092, & d = 2, \\ 0.053956, & d = 4. \end{cases}$$

The maximum transfers through the complete native and protected-memory attachments for arbitrary admitted native/reference inputs with this independently prepared work state. It is not an arbitrary work-input bound, and it excludes floating execution and physical service errors. The exact ideal logical-memory marginal below does not require this occupation bound.

Coherent native attachment. Let $L_h^{A,e}$ be the compiled native Kraus operator for law $A \in \{\mathsf{C},\mathsf{E}\}$, including its fair coin, and $K_r^{j(h)}$ the work branch including the first record's current-controlled kick. Source routing implements seam $(0,5)$ by exchanging port labels 1 and 5. With $e(0)=(0,5)$ and $e(1)=(0,1)$,

$$M_{h,r,s} = (L_s^{A,e(r_2)} L_h^{A,(0,1)}) \otimes K_r^{j(h)}, \qquad \sum_{h,r,s} M_{h,r,s}^\dagger M_{h,r,s} = I_{78} \otimes I_W.$$

Summing s, then r, then h proves completeness on arbitrary entangled native/work/reference inputs [51]. The execution keeps all 512 native amplitudes, padding and abort outcomes. Its complete terminal basis read has 295,106 outcomes for C and 820,166 for E. For input $(|\{2,3\}\rangle + e^{i\theta}|\{1,2\}\rangle)/\sqrt{2}$ and independent work, the declared native interference effect gives

$$p_C(\theta) = B + \frac{1+q}{8}\cos\theta, \qquad p_E(\theta) = B, \qquad q = \Pr(r_2 = 0 \mid j = 0).$$

Its C phase contrast is approximately 0.349153278 at $d=2$ and 0.351351028 at $d=4$, versus ideal zero for E. These distinguish two specified quantum instruments and are not independent empirical data.

The routing and memory decoder share a nine-qubit controlled-flip macro. Cyclic Gray ordering shares the commuting Walsh parities, reducing its source pulse count from 11,782 to 2,044: 512 phase pulses, 510 phase-exact CNOTs of three pulses each, and two target rotations. The scalar phase is retained and each parity ladder is restored. Exact parity checks and source replay preserve the entire 512-dimensional operator, including global phase, inverse and padding. This reduces the charged load/export events with no added qubits; it changes neither native repair nor the declared service assumptions [155, 51, 52].

Active encoded memory. Use the existing code

$$V|0\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}, \qquad V|1\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}.$$

Its flagged Knill–Laflamme identity is $V^\dagger L_h^\dagger L_h V = p_h I_2$, $p_h = k_h/4$, $k_h \in \{0,1,2\}$. The new source encoder and record-controlled full-register unitary R_h realize $R_h L_h V = \sqrt{p_h} V$ [52]. Correct after each native event, choosing final seam $(2,3)$ for $r_2 = 0$ and $(0,1)$ otherwise. Then

$$M_{h,r,s}^{\text{corrected}}(V \otimes I_W) = \sqrt{p_h p_s^{e(r_2)}}(V \otimes K_r^{j(h)}).$$

This operator identity preserves the logical/reference marginal after summing all work and record outputs, including initially entangled inputs. Conditional work measurements can steer that marginal within a branch. Any complete work-only instrument preserves this memory identity; accuracy of the specified work dynamics remains necessary for its full output claim.

The matched passive control omits recovery. For independent logical phase inputs and the same work preparation its complete plus/minus output distance is $1-\bar{q}$, where $q_j = \Pr(r_2 = 0 \mid j)$ and $\bar{q} = (q_{-1} + 2q_0 + q_1)/4$; every subsequent decoder therefore errs by at least $\bar{q}/2$ on one input.

The corrected distance is one. For the separate fixed seams $(0,1),(2,3)$, passive phase distance is zero and optimum Bell fidelity is $1/2$, while correction gives fidelity one. The optional decoded terminal read is destructive. Encoding, blank registers, source precision, storage and record services remain supplied; missing service errors are not zero. This is active protection of the stated qubit before readout, not arbitrary-carrier recovery, fault tolerance or an autonomous observer. Passive geometric and clock calibrations do not transfer to the corrected process.

6.7 Observer readouts and the fixed-reference clock

The readouts below extract observer representations, modular flow and regional structure from the same executed history [53]. Their quantum algebra, state pairing and access choices are premises, so the construction reconstructs the representation conditionally and derives neither the Born rule nor these premises from native repair.

Proposition 6.9 (Observer representations of the executed history)**.** *Let v_{ab} be the four unnormalized terminal vectors of the history at $d \in \{2,4\}$ and $\rho_{\mathrm{hist}} = \bigoplus_{a,b} |v_{ab}\rangle\langle v_{ab}|$ on $\bigoplus_{a,b} M_{4d^2}$.*

(i) For a positive functional ω the Gelfand-Naimark-Segal (GNS) construction $\langle [A],[B]\rangle = \omega(A^*B)$, $\pi(X)[A] = [XA]$, $\Omega = [I]$ has dimension dr on a block M_d whose density has rank r. The record and work algebra has GNS dimension $4(4d^2)$, that is 64 and 256; the four committed record projectors give a four-dimensional classical GNS space; the site-0 observer that forgets both records has GNS dimension d^2, that is 4 and 16; and the regional maps $J_R[A] = [\mathrm{embed}_R(A)]$ are isometric and intertwine the regional actions.

(ii) For faithful ρ_0 the modular group $\sigma_s(A) = \rho_0^{is} A \rho_0^{-is}$ moves number observables; the commutative record algebra has identity flow; states conditioned on the sharp terminal bit are nonfaithful on the full scalar algebra, with rank at most $d/2$.

(iii) At $d = 2$, $\rho_0 = \frac{1}{2}(I - \delta X)$ with $\delta = P(b=0) - P(b=1)$ and modular frequency $\nu = \log(P(b=0)/P(b=1))$. A source-compiled continuation of every terminal branch by $Z(\pi/4)$, $X(s\nu/2)$ and a fresh number pointer preserves the old two-bit record marginal and has readout law $P(1 \mid s) = \frac{1}{2}(m + \delta \sin(\nu s))$, where m is the total branch mass, equal to one for the normalized history.

(iv) Scalar regional restrictions are completely positive conditional expectations, normalized partial trace followed by identity embedding; they are unital and idempotent and in general not multiplicative. The centered components of the actual Hamiltonian give an interaction graph that is a path through scalar site 1, scalar site 0 and the shared charged-matter and link sector, with one edge from the scalar bond and one from the scalar and charge interaction.

Proof sketch. For $\rho = V \,\mathrm{diag}(\lambda) V^*$ of rank r the map $[A] \mapsto AV\sqrt{\mathrm{diag}(\lambda)}$ is the explicit GNS isometry [73]. The modular identities, including the finite Kubo-Martin-Schwinger (KMS) condition, are checked with this sign convention [74]. At $d = 2$, $-\log \rho_0$ equals a multiple of the identity plus $(\nu/2)X$, and the probe implements the Schrödinger dual of σ_s [53]. □

In the executed history $\nu \approx 0.5292248981159262$ and $\delta \approx 0.2586046272446501$ [71]. Ensemble frequencies read a modular interval on a declared monotone window, and full-cycle aliases are indistinguishable. Taking the post-measurement site-0 state as its own modular reference fails to reproduce the interacting dynamics. Its modular flow preserves it, while its density changes under H with derivative norms about 0.2086 at $d = 2$ and 0.6603 at $d = 4$, so no rescaling identifies the two flows on local expectations [59]. A fixed reference on the full work algebra removes this obstruction [54].

Proposition 6.10 (Fixed-reference modular relation)**.** *Let H be the finite interacting Hamiltonian on the work space of dimension $4d^2$, $\beta > 0$ and $\sigma_\beta = e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$. Then*

$$\sigma_\beta^{is} A \sigma_\beta^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \alpha_{-\beta s}(A), \qquad \alpha_t(A) = e^{itH} A e^{-itH},$$

so continuous Hamiltonian segments correspond to modular flow at $t = -\beta s$. For the full matrix algebra this identification of the two automorphism groups forces the Gibbs form of the reference. With record blocks $\Sigma_\beta = \bigoplus_r q_r \sigma_\beta$, $q_r > 0$, the weights cancel and every record projector is fixed. The actual state $\omega(t) = e^{-itH} \omega e^{itH}$ can change while σ_β is stationary, and $\operatorname{Tr}(\omega(t)E) = \operatorname{Tr}(\omega\, \sigma_{-t/\beta}(E))$ for every effect E.

Proof sketch. $\log \sigma_\beta = -\beta H - \log Z_\beta\, I$, and the scalar cancels in the conjugation. Equality of the generators requires $[\log \sigma + \beta H, A] = 0$ for every matrix A, and the commutant of the full matrix algebra consists of scalars [54, 78]. □

The reference, β and the full-work access are declared choices, and measurements and feedback act as interventions outside this one-parameter group. The receipt uses $\beta = 1$ and reference weights $(1/10, 2/10, 3/10, 4/10)$. Source continuations at model times 0, 1/2 and 1 append a gauge-number pointer to every post-sharp branch and retain all eight outcomes [54].

6.8 History repair, geometry, causal cones and field relations

Theorem 6.11 (Local repair of the completed history)**.** *Fix $d \in \{2, 4\}$, the native current j and the feedback setting. Let twelve event nodes carry unnormalized work vectors subject to the compatibility equations $x_{\mathrm{root}} = \mathrm{prepared}(j)$, $x_1 = U_1 x_{\mathrm{root}}$, $x_{(a)} = P_a x_1$, $y_{(a)} = F_a x_{(a)}$, $z_{(a)} = U_2 y_{(a)}$ and $w_{(a,b)} = B_b z_{(a)}$, of depths 0 to 5. Then:*

(i) exactly one assignment satisfies all equations;

(ii) the common fixed set of the local updates $x_v \leftarrow A_v x_{\mathrm{parent}(v)}$ equals that compatibility set;

(iii) every node-fair schedule reaches the compatible assignment from arbitrary initial vectors after finitely many updates and stays there; five covering rounds suffice, a forward topological pass needs eleven updates, and a reverse order may need 55;

(iv) with subtree weights w_v, equal to 11, 5, 4, 3 and 1 at depths one to five, the potential $\Phi = \sum_{v \neq \mathrm{root}} w_v \|x_v - A_v x_{\mathrm{parent}(v)}\|$ decreases by at least the repaired residual at every active ideal repair;

(v) every deterministic extraction from the completed assignment, including branch probabilities, terminal densities, GNS representations and modular parameters, is independent of the schedule.

Proof sketch. Agreement at depth zero forces agreement at each further depth. A node whose ancestors are correct becomes correct permanently at its next selection, and after the kth covering round every node of depth at most k is correct. Updating v zeroes its residual and changes each child's residual by at most $\|r_v\|$, since all maps are contractions [55]. □

The solved object is the complete branching instrument law for given apparatus inputs; its uniqueness selects no realized record pair, and the auxiliary solver is a host computation distinct from a native hardware process. The geometric and field statements that follow are dimensionless carrier displacements and event counts; physical position, momentum, length scale and propagation speed need their own operational identification.

Proposition 6.12 (Native geometric readout). *(i) Exact enumeration of all 4,680 state, seam and coin branches gives the one-attempt mean $T = I - L/60$ under uniform seam selection.*

(ii) With Q removing the constant mode and $C_n = QT^{2n}Q$, the normalized response $12C_n/\operatorname{Tr} C_n$ tends to $G = 4P_{\mathrm{slow}}$, where P_{slow} projects onto the rank-three band $5 - \sqrt{5}$; all eleven nonconstant modes are present at every finite attempt count.

(iii) For the readout $x(n) = \sum_p n_p u_p$ with frame directions u_p, a committed current j on the seam (a, b) gives the relative displacement $x(n') - x(n) = j(u_b - u_a)$, computable from the record without hidden loads.

(iv) The original controller spans two displacement directions. A declared continuation that sends the final scalar bit to $(0, 5)$ or $(0, 7)$ after the first event re-executes all 1,056 second events, and its three displacement directions $(2, 0, 0)$, $(1, 1-\varphi, \varphi)$, $(1, 1-\varphi, -\varphi)$, with $\varphi = (1+\sqrt{5})/2$, have determinant 4.

Proposition 6.13 (Native causal cones). *For a fixed seam schedule and shared coins, the load at a port v after the schedule is a function of the initial loads and coins in its backward dependency set. An initial difference supported on D influences v only along a time-ordered path, so $\operatorname{dist}(D, v) \leq$ edge crossings on that path $\leq$ elapsed seam events, and at most one edge is crossed per causal layer of disjoint simultaneous events.*

Exhaustive checks cover all 78 states for 27 schedule and coin pairs, and two executed examples show influence at graph distance two arriving only after two properly ordered seams [57]. Taking one edge and one causal layer as units gives native signal speed at most one. Units themselves are observer conventions [59]. If one model length, time and energy unit correspond to L metres, T seconds and E joules, and a common operational propagation law fixed a dimensionless speed c_{model}, choosing the speed of light of the International System of Units (SI) would impose only $L/T = c_{\mathrm{SI}}/c_{\mathrm{model}}$; an operational action constant $\hbar_{\mathrm{model}}$ would add $ET = \hbar_{\mathrm{SI}}/\hbar_{\mathrm{model}}$, and one overall scale would stay free. The construction does not supply c_{model} relating its native distance readout to its modular clock, so these relations calibrate none of the receipts. The causal-cone receipt is retained in the common-source package and belongs to no claim record.

Proposition 6.14 (Native field relations). *(i) Every event satisfies the discrete continuity law $x' - x = -b_e j$ with $b_e = \mathbf{e}_a - \mathbf{e}_b$.*

(ii) Conditional on the past and the selected seam, $\mathbb{E}[j] = d/2$ and $\operatorname{Var}(j) = (|d| \bmod 2)/4$.

(iii) Under the uniform calibration the mean load obeys $\mu_{k+1} - \mu_k = -L\mu_k/60$, exactly, by $\mathcal{K}X = XT$ for the 78-state kernel $\mathcal{K}$ and load matrix X; the deviation $V = \frac{1}{2}\|\mu - \frac{1}{6}\mathbf{1}\|^2$ decreases strictly unless the mean is constant, while every binary trajectory keeps $\|x - \frac{1}{6}\mathbf{1}\|^2 = 5/3$.

(iv) The slow band has mean multiplier $q = (55+\sqrt{5})/60 \approx 0.953934466291663$, every nonconstant multiplier lies in $(0, 1)$, and the second-difference coefficient $q + 1/q - 2 > 0$ has the opposite sign to the coefficient $-4\sin^2(\omega\Delta t/2) \leq 0$ of an undamped oscillator; no positive rescaling of the attempt count turns this closed mean law into a wave law.

Proof sketch of Propositions 6.12 to 6.14. The fair coin averages a seam to its endpoint mean, which gives the mean operator and the conditional current moments; the other two positive response eigenvalues have strictly smaller magnitude than $1 - (5 - \sqrt{5})/60$, which gives the limit in Proposition 6.12(ii). Conservation gives the displacement identity. For the causal cones, scanning the schedule backwards and adding both endpoints of every seam that touches the dependency set proves the dependency claim by induction [56, 57, 58]. □

A common continuum operator. On the declared spherical source family, the native cell Green operators converge in norm along one common subsequence to a positive compact injective operator G on the mean-zero observation space. Its inverse $A = G^{-1}$ is densely defined and self-adjoint. Restoring the constant zero mode gives a conservative strongly local regular Dirichlet form, with domain H^1 on the supplied flat-polyhedron comparison surface. Its energy is bounded above and below by positive multiples of the gradient energy, and its heat-trace spectral dimension is two. Full native density correlations converge to $e^{-\tau A}$, while the admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. The full-space extension $\bar{A} = A \oplus 0$ satisfies

$$\|\mathbf{1}_F e^{-\tau \bar{A}} \mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)}.$$

Its cosine propagator vanishes between these supports when $|t| < \sqrt{2}\, d(E,F)$. The distance is the declared flat comparison distance; neither it nor the coherent parameter has a laboratory calibration. Two consecutive long observation blocks of the same native history supply a continuous classical Gaussian wave readout with covariance $G\cos((t-u)\sqrt{A})$ in $C([0,T], H^{-2}(S^2))$ [11]. Observation count tends to infinity before spatial refinement. The wave readout is declared processing, distinct from passive density waiting and from the quantum vacuum. Neither a unique whole-sequence limit nor an observed $1+3$ spacetime is proved.

Actual native density histories. At level l, let $V_l = 240\,4^l$, $q_l = V_l/3$ and $\chi_l = q_l(V_l - q_l)/[V_l(V_l-1)]$. In the uniform canonical equilibrium, with IID uniform seams and fair repair, repeat each mean-zero test function's cell averages on its twelve native ports. The actual snapshot readout

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\, 16^l \tau \rfloor}(p) - 1/3]}{\sqrt{V_l \chi_l}}$$

converges jointly at every finite collection of times to a stationary Gaussian family with covariance $\langle f, e^{-|\tau-u|A} g\rangle$. This limit refines space and native attempt count simultaneously. To establish the joint law, represent each unread repair as a fair identity-or-swap. The conditional profile overlaps concentrate with variance at most $4\|g\|_\infty^2 \|f\|_{\ell^2}^2 / V_l^2$, uniformly in attempt count; a conditional finite-population central limit theorem then applies. The equal-time covariance is the identity. The declared spatial filter $G_l^{1/2}$ gives limiting covariance $Ge^{-|\tau-u|A}$. The Gaussian law is reflection positive: past and future are conditionally independent given the time-zero field. Its reconstructed one-particle semigroup is $e^{-\tau A}$, with generator A, not $\sqrt{A}$. These are finite-dimensional process laws; no path-space tightness, coherent quantum preparation or laboratory duration is inferred [11].

The reference-state boundary. Increasing native-return caps match modular phase magnitudes to the cosine limit; a fixed cap freezes the low-band phase. On the full configuration algebra, however, the uncapped reference approaches normalized trace with half-trace error at most $3/(8\,2^l)$. Its density-wave sector has vanishing weight. Fixed-parameter modular evolution becomes trivial in normalized-trace L^2, and bounded-overlap fluctuation commutator expectations vanish. These bounds do not exclude refinement-rescaled time, other states or other fluctuation regimes. If the relevant fluctuation moments converge, any limiting scalar canonical commutator in this bounded-overlap regime must be zero; a quantum vacuum is not identified [11].

7 A prediction protocol

A response without a fitted coefficient. In the stationary binary number sector, disclose local occupations and apply one extra fair native repair on seam (a,b) exactly when $n_a = 1$. Detailed

balance gives, for any fixed future record event F whose continuation does not consult the earlier pulse mark,

$$\Pr_{\text{feedback}}(F) - \Pr_{\text{passive}}(F) = \tfrac{1}{2}\mathbb{E}_{\text{passive}}[(n_b - n_a)\mathbf{1}_F].$$

Separate likelihood weights retain every trigger, coin and current label; the intervention contributes to each incident clock. Complete capped clock/field forecasts and the native spatial Green response satisfy this identity exactly [11]. The forecasts have no matched public intervention dataset and establish no discriminator against SM+GR.

A controlled continuum response. For a disclosed port profile h, a uniform seam (a, b) executes its unchanged fair repair with probability $[1+\epsilon(h_b-h_a)(n_a-n_b)]/2$. The clock counts executed repairs, including zero currents; selection slots and controller services are charged separately. Its exact linear occupation response is $2\chi K(I - K^r)h$, with the extra K arising from state-dependent clock stopping. The normalized coordinate profile on the refining source gives

$$\partial_\epsilon \mathbb{E}\eta_l(g)|_0 \longrightarrow 2\langle g, (I - e^{-\tau A})f\rangle.$$

At $\tau = 1$, a cap of four selection slots per target repair retains actual timeout fields. Stopped likelihood bounds control the nonlinear mean error by $36\epsilon^2\|g\|$. For a unit-axis flat coordinate and $\epsilon = 10^{-6}$, $\liminf_l \mathbb{E}_\epsilon \eta_l(f) \geq \epsilon\, 9/32792 - 36\epsilon^2 > 0$. Finite fair-bit activation approximations have vanishing terminal-mean error with explicit service counts. This is a controlled signal on the same continuum, without a unique nonlinear limit or an identified physical force [11].

7.1 The phase experiment

The classical repair law fixes the diagonal of every admissible record law and none of its coherences (Theorem 2.3). A fixed interference experiment reads the multiplier g that separates the members, and its count protocol, which compares the executed laws C and E, is fixed before any data [38, 39]. All members are ordinary quantum instruments, so the test separates record laws and makes no claim to separate OPH from quantum mechanics.

Write $x = \{2, 3\}$, $z = \{1, 2\}$ and $w = \{0, 2\}$ for binary configurations on the 78-dimensional carrier. Prepare $|\phi_\theta\rangle = (|x\rangle + e^{i\theta}|z\rangle)/\sqrt{2}$ for $\theta \in \{0, \pi, \pi/2\}$, or the incoherent mixture $\frac{1}{2}(|x\rangle\langle x| + |z\rangle\langle z|)$. Apply one repair on the seam $(0, 1)$ and retain its record. Then measure $P_+ = |\phi_0\rangle\langle\phi_0|$ and $P_- = I - P_+$ inside each committed record block.

Proposition 7.1 (Phase law of the two record instruments)**.** *For the preparations above:*

(i) the nonzero recorded source densities of C *are* $\frac{1}{2}|\phi_\theta\rangle\langle\phi_\theta|$ *for* $(c, j) = (0, 0)$, $\frac{1}{4}|x\rangle\langle x|$ *for* $(1, 0)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*; those of* E *are* $\frac{1}{4}|x\rangle\langle x|$ *for* $(c, d) = (0, 0)$ *and* $(1, 0)$, $\frac{1}{4}|z\rangle\langle z|$ *for* $(0, -1)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*;*

(ii) the plus probabilities are $p_{\mathsf{C}}(\theta) = \frac{3}{8} + \frac{1}{4}\cos\theta$ *and* $p_{\mathsf{E}}(\theta) = \frac{3}{8}$*, and the incoherent preparation gives* $3/8$ *for both; the phase-reversal contrast is* $1/2$ *for* C *and zero for* E*;*

(iii) all four preparations have identical classical load probabilities, the readout preserves the probability of every old record, and no record is erased and no outcome postselected.

Proof sketch. On the seam $(0, 1)$ the configuration x has contrast zero for both coins, and z has contrast -1, current 0 for coin zero and current -1 for coin one, which moves it to w. The coherent sum survives only in the C branch $(0, 0)$, where both currents vanish; E separates x from z by their contrasts. Summing $\mathrm{Tr}(P_+\rho_r P_+)$ over the branches gives (ii) [39]. □

Table 8: Plus probabilities of the four fixed settings [38, 14].

Preparation	C_0	C	E	T
Phase 0	$\frac{3}{8}+\frac{\sqrt{2}}{4}$	5/8	3/8	3/8
Phase π	$\frac{3}{8}-\frac{\sqrt{2}}{4}$	1/8	3/8	3/8
Phase $\pi/2$	3/8	3/8	3/8	3/8
Incoherent mixture of x and z	3/8	3/8	3/8	3/8

For every admissible record law the plus probability is

$$p_\Phi(\theta) = \tfrac{3}{8} + \tfrac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta}).$$

The empty seam of x stays in place, and z waits with probability $\frac{1}{2}$ while keeping coherence g with x. Proposition 7.1 is the case $g = \frac{1}{2}$ for C and $g = 0$ for E. The aggregate plus frequencies of the settings 0, π and $\pi/2$ therefore estimate $\mathrm{Re}\,g$ and $\mathrm{Im}\,g$ for whichever law is realized [14]. Laws with equal g, such as E and T, are separated by the same construction with the full seam $\{0,1\}$ in place of z, which gives $\frac{1}{2} + \frac{1}{2}\,\mathrm{Re}(k\,e^{-i\theta})$.

The count protocol fixes four settings with 8,192 fresh independent attempted trials each and no optional stopping. Each attempt contributes to one of twelve common bins of coin, current and outcome, or to an abort; every frequency uses all 8,192 attempts. At each setting it scores the twelve bin indicators, the plus indicator and the abort indicator, $m = 4(12+2) = 56$ statistics in all. The sampling half-width is $b = 1/32$, the independently established apparatus probability budget is $\eta = 1/64$, and the familywise sampling requirement is $\delta = 0.001$.

Proposition 7.2 (Sampling certificate)**.** *If the attempted trials of each setting are independent, then with $n = 8192$ and $b = 1/32$ the probability that any of the 56 observation intervals $[\hat{p}-b, \hat{p}+b]$ misses its actual probability is at most $112e^{-16} < 112/S_{32} < 0.000012606 < \delta$, where $S_{32} = \sum_{k=0}^{32} 16^k/k!$.*

Proof. Hoeffding's inequality [81] gives $2e^{-2nb^2} = 2e^{-16}$ for each statistic, and a union bound over 56 statistics gives $112e^{-16}$. Positivity of the exponential series gives $S_{32} < e^{16}$, and the final comparison is exact rational arithmetic [39]. □

A candidate is rejected when some observation interval is disjoint from its ideal probability widened by η. If the apparatus contract holds, rejection can occur only when an observation interval fails, which proves the familywise bound; both candidates use the same intervals, so testing both needs no further union bound. If the calibration guarantee itself fails with probability at most β, the combined bound is $112/S_{32} + \beta$. An abort counts in the denominator, has ideal probability zero, and a statistically resolved excess over its allowance returns the verdict that the apparatus budget is exceeded. Without supplied counts the verdict is "no data". Software demonstrations are labelled synthetic and external count files unverified, and every result reports empirical confirmation as false [38]. If one candidate survives and the other is rejected, the counts discriminate the conditional instruments; if both survive, the test does not separate them; if both are rejected, neither specified apparatus model explains the counts within the declared premises. In every case the same counts bound the realized coherence g.

7.2 The source-executed implementation

Both complete recorded instruments, the four preparations and the readout also run as emitted twelve-port control words [36, 37]. The encoding is a mathematical isometry; a physical interface

from occupation states to the code is not established. A configuration is written as the seam pair and the ten spectators. The six endpoint pairs use three source bits and the spectators six bits, which places the 78 configurations injectively in a nine-qubit register with 434 unused addresses. A local permutation with blank record bits realizes each repair from phase-exact transpositions, and the fair placement coin is itself prepared by a source pulse and recorded.

Proposition 7.3 (Exact source realization of the two instruments)**.** *In the nine-qubit encoding J:*

(i) the encoded recorded channels intertwine the native ones,

$$\mathcal{E}_{\mathrm{code}}(J\rho J^*) = (I_{\mathrm{record}} \otimes J)\, \mathcal{E}_{\mathrm{native}}(\rho)\, (I_{\mathrm{record}} \otimes J^*),$$

on all 6,084 matrix units of the 78-state carrier and therefore with every untouched reference; the complete Gram identity has 156 nonzero Kraus entries;

(ii) E *is* C *followed by a three-pulse source controlled-NOT (CNOT) that copies the seam-total parity p, with contrast decoded as $d = 2j + (2c-1)p$; relative to the retained coin and current labels one additional binary record is necessary and sufficient, and forgetting it leaves parity dephasing without restoring the coherence of* C*;*

(iii) the readout is the Lüders instrument [85] of the rank-one plus projector and its complement, realized by coupling the plus projector to a blank pointer and measuring only the pointer; the minus outcome is one rank-77 projector whose internal coherences survive, and reading the full computational basis would implement a different instrument.

Proof sketch. For fixed current the endpoint map is an injective translation, and for fixed contrast the spectators and endpoint sum reconstruct the input, so every input column of the joint coin, record and source isometry has two entries $2^{-1/2}$ with disjoint supports. An independent floor and ceiling enumeration checks the permutation images and all 6,084 matrix units for each candidate. For (ii), the inputs $\{2,3\}$ and $\{1,2\}$ share the C record $(0,0)$ and have contrasts 0 and -1, so a single extra value cannot separate them [37]. □

The complete attempted trial costs 32,006 native pulse requests for the pure C settings, 32,007 for the diagonal C setting, 32,009 for the pure E settings and 32,010 for the diagonal E setting, with one proper-code load and one export charged per pulse. Each readout costs 26,632 pulses, a local C repair 5,370 and a local E repair 5,373; the peak live code width is 11 qubits for C and 12 for E [36]. Executed terminal states reproduce the probabilities of Table 8 and the full output densities, and the synthetic counts, 8,192 attempted trials per setting drawn from executed branch norms, feed the unchanged count protocol. For each synthetic dataset the evaluator keeps its generating candidate and rejects the other. A compositional error contract,

$$N_p u + N_l\, l + N_e\, e + p + z + c + a + r + s + m + g + n \le \frac{1}{64},$$

with pulse, load and export counts N_p, N_l, N_e, their uniform channel errors u, l, e, and whole-trial allowances for preparation, initialization, coin, randomizer, records, storage, measurement, coefficient compilation and numerical execution, gives sufficient conditions for the protocol's budget $\eta = 1/64$; an equal allocation at the largest schedule is $1/6146496$ per term. The same source alphabet, code services and classical transitions thus admit two implemented quantum instruments whose common observable at phase zero differs by $1/4$, so these ingredients alone do not select the quantum record law.

7.3 Retrospective audit of the inherited coupling output

The detector of Section 4.3 evaluates the inherited structured-running branch of the closure map. The flagship synthesis reports a second, self-consistent closure map with a much smaller residual [2]; this construction consumes the bare source map only. A retrospective diagnostic compares that output with the 2022 recommended value of the Committee on Data of the International Science Council (CODATA)

$$\alpha^{-1} = 137.035999177(21)$$

[77]. The inherited output, approximately 136.994835177413, falls short of the central value by about 300.388218 parts per million, and exact rational arithmetic shows its narrow numerical interval disjoint from the band of five reported standard uncertainties around the central value [38, 39]. The band is a declared comparison allowance without the status of a confidence interval. The audit rejects the additional hypothesis that this output is the complete physical endpoint with zero missing correction; the documented incomplete-transport contract of the branch requires physical transport work. No fit is performed, P is unchanged, and no theory significance is assigned.

8 The premise map

The premise map relates the construction to the conditional theorems of the library [16, 17]. It is a navigation document generated from a machine-readable crosswalk, and it applies one rule. A result "H implies C" discharges an input I only when C establishes I on the same domain with the required errors and H has independent support; a change of regulator, state or code, or a calibration relabelled as a derivation, does not discharge an obligation. Accordingly the map distinguishes two kinds of entry. An antecedent can be discharged for the finite objects of the construction by execution or proof, or it can be supplied. Discharge on the finite domain is a different statement from discharge of the corresponding physical premise on a physical refinement tower. Five of the six claim records of Table 2 consume no canonical physical premise; the detector record consumes the accepted closure inputs P and N (P numerically, N as the carrier bound of Section 2.6) and the selected coupling map, and lists the same-quantity bridge for P and the physical Thomson readback contract as unsupplied [1]. The current crosswalk represents all 84 canonical premise identifiers in 27 groups and inventories 339 registered claims and 660 Lean modules. This inventory does not individually instantiate or prove every listed claim, ordinary assumption or imported result in the observer model [16]. Table 9 summarizes seven theorem routes.

Finite entropy and the gravitational premise. For positive sector counts d_i, normalized probabilities p_i and fixed $M = \sum_i d_i$, the counting entropy obeys

$$S_{\mathrm{count}} = H(p) + \sum_i p_i \log d_i = \log M - D(p\|d/M) \le \log M.$$

The maximizer is $p_i = d_i/M$. In the positive-real relaxation, joint fixed-M variations have zero first derivative there and second derivative $-\sum_i (p_i' - d_i'/M)^2/p_i$. This is not an operation varying integer dimensions. If sectors contain bulk factors, full entropy saturation also requires maximally mixed bulk blocks and counts their dimensions. Uniform native preparation realizes counting weights; consensus against an arbitrary reference need not do so. Fixed count has not been identified with fixed physical volume, nor the logarithmic sector term with a geometric area term of universal coefficient [12]. For every cap of the wave reference's return-length law, an absolute bound holds

on the full $D = \binom{V}{V/3}$ configuration space:

$$\Delta = \log D - S(\sigma) \le \frac{2025}{64V}, \qquad I_\sigma(R : R^c) \le \log J + \Delta,$$

where J counts the allowed regional particle numbers. A complete base-face patch has $V/20$ ports and $3\,2^l$ crossing seams. Its stationary reduced entropy is extensive in port count, while the return reference's mutual information divided by crossing-seam count vanishes. Neither observable therefore has a nonzero universal coefficient times that boundary count. This does not exclude interpreting a whole spherical screen as a bulk boundary; that physical region map is a premise [12].

Four retained audit controls fix exact boundaries of the construction [13]. Each rejects a named identification or an overly broad inference; none rejects the theorem premises or requires a unique designed implementation.

Complete-carrier center. The centers of M_{66} and M_{78} consist of scalars, so the unextended working algebra cannot supply twelve nonzero primitive central ports. Binary occupations are noncentral, overlap and sum to $2I$, and total-two occupations also admit the eigenvalue two. The native matrix algebra is therefore kept as a sector; a separate twelve-block central-address extension supplies a central algebra with port-controlled operations, and occupation measurement is a separate contract.

Marked triplets. The slow Gram matrix satisfies $G^2 = 4G$, $\operatorname{Tr} G = 12$ and $(5I - A)G = (5 - \sqrt{5})G$. One fixed order-five symmetry has trace $(1 - \sqrt{5})/2$ on the slow frame and $(1 + \sqrt{5})/2$ on the lower response triplet, which excludes an invertible intertwiner for the same action. Spatial and internal response domains are therefore kept distinct; a whole-law Galois substitution or an explicit twist of the action needs its own attachment.

Feedback class. With the two endpoint loads in the uniform mixture of their one-particle configurations, a rational number-preserving exchange gives endpoint probabilities 769/1250 and 481/1250 for the two private preparations, a difference of 144/625. Four Kraus operators certify that it is CPTP, and its private commutant has dimension one, which excludes every single-product exponential representation, alternate logarithm branches included. The no-feedback theorems for reduced loads and single products hold within their stated classes, and this channel marks the class boundary.

Configuration equilibrium. For $A = I + L_{\text{path3}}$ the controls $\varepsilon = \pm 1/16$ keep the normalized configuration Gibbs marginal and change the full reference and the vacuum. For a stiffness response $Y = \gamma(A + \varepsilon A^2)$ one has $Y_{13} = \gamma\varepsilon$ and a third-mode fit residual $6\gamma\varepsilon$, while the actual acceleration is $-Yq$. The theorem that identifies inertia with the same capacity keeps that premise, and equal equilibrium configurations do not select the kinetic law.

9 Open connections

The primitive selection obligations are the carrier architecture, the observer interface and its implementation, and the complete repair rule [25]. Effective fields, transport, clock relations and interactions must follow from that same process. The scalar and matter actions supply conditional

Table 9: Theorem routes, antecedents discharged on the declared construction, and supplied antecedents [16, 13].

Route	**Discharged by execution or proof**	**Supplied**
Consensus and repair [5]	Committed-record preservation for the native and common-source instruments; completion identities and the exact gap of the record-discarded channels; schedule-independent recovery of the completed common-source history with a strictly decreasing residual.	Repair completeness, local diamonds and well-founded transactional descent for an unrestricted observer menu; conflicting updates; federation; global history repair executed by native hardware.
Quantum records [6]	All-input CPTP maps with coherent write isometries; complete binary instruments retaining both outcomes; the exact correctable algebra, optimal recovery and source-executed active encoded memory; the compiled joint native/work channel on arbitrary referenced inputs; GNS representations of the actual observation state.	Hilbert representation and canonical relations, subsystem split, Born readout, coherent preparation and readout services, a physical encoding of loads in the code.
Record capacity [6]	M_0 for each declared record interface with a carrier projector certificate; the bound $\ln M_0 \leq N$ on a declared carrier.	A capacity-indexed source family, complete-fiber closure, a physical selector, and the same-carrier horizon identification behind $\Lambda \ell_\star^2 = 3\pi/N$.
Gauge structure [7]	Branch charge covariance of the designed assignment $Q = g(P)N$; a four-state CAR and link code with operator Gauss constraints and even parity under source execution.	Physical current and action for the complete response algebra, the same-source category, the global form, coupling attachment, nonabelian and chiral matter.
Scalar continuum [6, 2]	Refinement transfer to complete histories and native feedback; occupation limits with complete-output bounds for free and interacting two-site actions; the infinite-occupation interacting signal. A native regular local H^1 form of spectral dimension two governs spherical Gaussian fields, decay and admitted finite-speed comparison waves [11].	The scalar action, quantization and vacuum; inertia; the model clock; a spatially refining family for the interacting process.
Modular clocks [8]	Finite modular flow of actual observation states; a programmed modular clock with a readable law; the fixed-reference relation $t = -\beta s$ and a gauge probe on the continued smooth history. A specified native-return reference links rescaled wave phases to a modular generator; its full-state limit is tracial at fixed modular parameter, and fixed caps freeze the low-band phase [11].	Selection of a physical reference and its complete modular dynamics; Bisognano-Wichmann identification; common null translations; physical clock calibration; the relation to causal depth.
Gravity [8]	A rank-three response attached to committed records and native causal cones; a common local spherical continuum and continuous observation and classical-wave limits under their count-first contract [11]. These supply no responsive metric or physical stress.	Physical region, metric and area attachment; geometric identification of finite counting-entropy stationarity; small-ball geometry; conserved stress; universal coupling and normalization on one inhabited physical tower.

comparison constructions; placing a target physical law inside an interface or update would not derive it. The connections below test the target of Section 3 without requiring a uniqueness theorem for the chosen design.

A spatially refining family. A family of source programs that refines in space, carries the charges, interactions and instruments of Section 6, retains its records, and bounds numerical and physical services along the way would connect the construction to a continuum. Two oscillators at increasing occupation cutoff stay two oscillators, and the spatial bounds of Section 5.1 transfer to another implementation only through an explicit map. No continuum limit of the interacting family is established.

Matter, geometry and clocks on one process. Interacting matter, responsive geometry and a physical clock calibration have to attach to the same state, event law and retained records. This needs stress and geometric readouts from the same family, the balance and Ward identities with controlled tails, a calibrated scale, a choice of modular reference produced by the construction, and a relation between modular readings and causal depth. The calibration examples show that fixing the speed of light relates length and time units and leaves an energy scale free [59].

Physical realization of the phase test. The test of Section 7 needs a realized preparation and observable, an independently established service budget, and complete records collected under the fixed protocol. Its outcome is a measured coherence g, which locates the realized record law within the family of Theorem 2.3. A physical model of nature additionally needs sector-specific predictions; comparisons with data seen before a protocol was fixed are postdictions.

Joins between executed components. Several certified components run on separate schedules with separate error budgets. The compiled native instruments join the four-read smooth common source and active encoded memory (Section 6.6); their physical services remain uncalibrated. The two-read free-field history of Section 5.2 has no source compilation; the local split-update engine and the two-read continuation of Section 5 carry different bounds; and the modular and fixed-reference clocks of Section 6.7 continue the finite sharp history without an occupation limit of their own. The smooth continuation bounds its actual later gauge probe, without proving convergence of Gibbs modular data. Each remaining join needs its own implementation and error analysis, and certified numerical execution together with independently established physical service bounds is a separate budget for every executed word. The smooth and sharp common-source instruments are compiled only at $d = 2$ and $d = 4$; larger cutoffs are analytic comparisons.

Longer observers. Finite composed histories and active encoded memory have explicit continuation and resource contracts. Extending them to autonomous operation requires accounted fresh records, storage, resets and physical services while preserving the demonstrated behaviour. The incident-tick memory test becomes an implemented apparatus once the N4 preparation, occupation detector and local event counter are joined; and a native quantum realization of the global history repair of Section 6.8 is a separate construction requirement.

10 Reproducibility

Every executable result has a replayable receipt. Package manifests bind source, document and receipt artifacts by SHA-256. The repository's verification entry point recomputes these hashes,

replays retained receipts and compares canonical bytes. Exact certificates must match exactly, and floating diagnostics carry declared replay tolerances. Independent test suites check the executable statements against separately written oracles, including adversarial mutations. Analytic limit proofs state their domains, normalizations and order of limits; a finite receipt does not replace those proofs. Model sources and the premise crosswalk are available at the pinned revisions cited with their results [11, 16]; the flagship synthesis and basis document are cited on the main branch [2, 3].

Statements and declarations

Funding. The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.
Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.
Data availability. The model documents, executable sources, retained receipts and claim records supporting this study are available in the public OPH repository at the baseline revision [1] and the revisions cited with individual results.
Corresponding author. Correspondence should be addressed to Bernhard Mueller, bernhard@floatingpragma.ai.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Aharonov_Bohm_loop_phase [72].
- analytic_vector_self_adjointness_and_core_strong_resolvent_convergence [131, 140].
- binary_Kraus_Stinespring_instrument [84, 146, 135].
- Bose_Hubbard_lattice_action [115].
- canonical_lattice_scalar_quantum_simulation [117].
- Caputo_Liggett_Richthammer_exclusion_spectral_gap [75].
- classical_CHSH_local_response_bound_and_published_Bell_null_contract [100, 110].
- classical_controlled_quantum_channels [153].
- classical_Weyl_eigenvalue_counting_law [141].
- coherent_state_evolution_under_positive_quadratic_action [106, 101].
- coisometry_involution_quantum_walk_spectral_mapping [148, 111].
- conditional_Hoeffding_Azuma_concentration_for_randomized_Gram_witnesses [81, 94].
- conditional_law_sequential_coupling [123].
- conditional_probability_and_Brier_decomposition [96, 130].
- convex_Schrodinger_spectral_gap_comparison [92].
- Davies_Gaffney_finite_wave_propagation_equivalence [102].
- Doob_L2_martingale_maximal_inequality [91].
- exact_rational_arithmetic_with_directed_root_and_pi_enclosures [129, 142].
- finite_acyclic_dependency_induction_and_contractive_residual_inequality [118, 95].
- finite_CAR_representation_and_gauge_invariant_sector_isometry [116, 74, 121].
- finite_Gibbs_reference_modular_Hamiltonian_identity [109, 74].
- finite_Kraus_Stinespring_construction [146, 122, 99].
- finite_Lindblad_master_equation [107, 126].
- finite_matrix_block_elimination_and_stopped_instrument_resolvents [157, 133].

- finite_matrix_conditional_expectations_and_dual_fixed_observables [152, 93].
- finite_matrix_GNS_representation_and_modular_automorphisms [73, 150].
- finite_state_martingale_functional_central_limit_theorem [156].
- finite_stationary_endpoint_correlations [123].
- first_hit_channel_geometric_series [120].
- Gauss_Hermite_quadrature_remainder [149, 134].
- Gaussian_Green_covariance_and_generalized_free_field_construction [145].
- Gaussian_Weyl_characteristic_function [112, 154].
- Hajek_finite_population_central_limit_theorem [124].
- Hermite_half_line_projector_matrix_elements [149, 134].
- Hoeffding_finite_sample_concentration_and_union_bound [81].
- number_moment_Gronwall_and_projected_Duhamel_comparison [108, 137].
- Poisson_factorial_moments_and_finite_dependency_cones [104, 138].
- published_trapped_ion_anharmonic_dynamics [128].
- published_trapped_ion_preparation_and_characteristic_readout [128].
- quantum_relative_entropy_data_processing [125].
- Ramsey_binary_quantum_instrument [139].
- Richardson_extrapolation [147].
- round_sphere_finite_perimeter_isoperimetric_inequality [97].
- Schur_complement_positive_resolvent_bound [157, 113].
- spectral_projector_strong_convergence_at_nonatomic_endpoints [87, 119].
- spectral_theorem_for_self_adjoint_operators [87, 119].
- spherical_harmonic_orthogonality_and_Laplacian_spectrum [134].
- Steklov_boundary_Weyl_law [105].
- strong_resolvent_and_spectral_projection_convergence_at_continuity_points [87, 119].
- symmetric_product_formula_integral_Taylor_remainder [114, 98].
- tensor_product_CP_conditional_expectations [151, 136].
- total_variation_contraction_under_finite_classical_continuation [103, 123].
- trace_distance_contraction_and_unitary_channel_composition [143, 132].
- Walsh_Gray_diagonal_unitary_synthesis [155].
- zero_error_independence_number [144, 127].

References

[1] FloatingPragma (2026). *Observer Patch Holography*: papers, Lean library, executable model, receipts and claim registries. Public repository; the model documents and receipts are cited at the revision below, at which the components were certified, and the flagship synthesis, the basis document, the premise reference and the generated premise map on the main branch. https://github.com/FloatingPragma/observer-patch-holography/tree/196b271021967927bc2626283b6cf7955972a4b2

[2] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., Nguyen, P., Kim, J., Matscheko, D., Hill, J., Glynn, W. T., Visser, M. A., Anirudha, K. A., and de La Fournière, B. (2026). *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/main/flagship/from_observer_consensus_to_standard_physics.pdf

[3] FloatingPragma (2026). *The OPH basis*. Basis document: the closure hypothesis, the no-outside rule, the agreement condition and the executable specification. https://github.com/FloatingPragma/observer-patch-holography/blob/main/docs/BASIS.md

[4] FloatingPragma (2026). *OPH premise reference*. Formal statements of the theorem-library premises A1, A2 and A3. https://github.com/FloatingPragma/observer-patch-holography/blob/main/docs/AXIOM_REFERENCE.md

[5] Mueller, B., Xue, K., Kim, J., Anirudha, K. A., Matscheko, D., and Hill, J. (2026). *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/b2e8dd0537593e6e08f8eb6e2bc8f3b9065454c7/paper/reality_as_consensus_protocol.pdf

[6] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., and Nguyen, P. (2026). *Observers Are All You Need*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/b2e8dd0537593e6e08f8eb6e2bc8f3b9065454c7/paper/observers_are_all_you_need.pdf

[7] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., and Matscheko, D. (2026). *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/b2e8dd0537593e6e08f8eb6e2bc8f3b9065454c7/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[8] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Visser, M. A., Matscheko, D., and de La Fournière, B. (2026). *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/b2e8dd0537593e6e08f8eb6e2bc8f3b9065454c7/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[9] Mueller, B., Osika, A., Xue, K., Cassie, B., and de La Fournière, B. (2026). *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/b2e8dd0537593e6e08f8eb6e2bc8f3b9065454c7/paper/screen_microphysics_and_observer_synchronization.pdf

[10] FloatingPragma (2026). *The runnable OPH model*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/README.md

[11] FloatingPragma (2026). *A common native continuum operator, wave observations, modular phases and repair response*. Executable model, analytic bounds and retained source receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/47aba948d4c0acad7092759fc223eae5462b4cba/model/README.md

[12] FloatingPragma (2026). *Joint counting stationarity and its geometric boundary*. Exact entropy identity and finite source controls. https://github.com/FloatingPragma/observer-patch-holography/blob/47aba948d4c0acad7092759fc223eae5462b4cba/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[13] FloatingPragma (2026). *OPH model construction from local repair*. Construction guide with completion target, acceptance criteria and audit controls. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/CONSTRUCTION.md

[14] FloatingPragma (2026). *Admissible record laws for native repair*. Model document and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/03c26e8aaccfd6e7cd85e32214262e7d6a518447/model/record_laws/README.md

[15] FloatingPragma (2026). *Concrete repair-law candidates*. Candidate register. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/CANDIDATES.md

[16] FloatingPragma (2026). *OPH reverse-engineering table: model and theorem premises*. Generated premise map and concrete theorem bindings. https://github.com/FloatingPragma/observer-patch-holography/blob/47aba948d4c0acad7092759fc223eae5462b4cba/model/PREMISES.md

[17] FloatingPragma (2026). *Model and theorem navigation map*. Machine-readable input of the premise map. https://github.com/FloatingPragma/observer-patch-holography/blob/b2e8dd0537593e6e08f8eb6e2bc8f3b9065454c7/tracking/model_theorem_map.json

[18] FloatingPragma (2026). *OPH interpretation and working terms*. https://github.com/FloatingPragma/observer-patch-holography/blob/b2e8dd0537593e6e08f8eb6e2bc8f3b9065454c7/docs/INTERPRETATION_AND_TERMS.md

[19] FloatingPragma (2026). *Runnable finite observers*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/README.md

[20] FloatingPragma (2026). *Finite current-record controller: mathematical account*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/DERIVATION.md

[21] FloatingPragma (2026). *Native observer records reuse the existing capacity construction*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/CAPACITY.md

[22] FloatingPragma (2026). *Quantum mixing of the native C/E repairs*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/QUANTUM_MIXING.md

[23] FloatingPragma (2026). *Recovery with complete native records*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/RECORD_RECOVERY.md

[24] FloatingPragma (2026). *Coherent continuation: native attachment and nonselection*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/3d1846b017a109525902c42a35134dcbad826277/model/observer/COHERENT_CONTINUATION.md

[25] FloatingPragma (2026). *Selecting one source, interface and repair law*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/3d1846b017a109525902c42a35134dcbad826277/model/SOURCE_SELECTION.md

[26] FloatingPragma (2026). *P-dependent electromagnetic observation and native feedback*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/p_response/README.md

[27] FloatingPragma (2026). *Charged realization, binary instrument and feedback*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/p_response/DERIVATION.md

[28] FloatingPragma (2026). *Detector access and the conditional capacity budget*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/p_response/ACCESS_CAPACITY.md

[29] FloatingPragma (2026). *Validation of the native-record electromagnetic adapter*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/p_response/VALIDATION.md

[30] FloatingPragma (2026). *A scalar detector inside the native observer protocol*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/scalar_response/README.md

[31] FloatingPragma (2026). *A native record coupled to a scalar field instrument*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/scalar_response/DERIVATION.md

[32] FloatingPragma (2026). *Conditional detector prediction*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/scalar_response/PREDICTION.md

[33] FloatingPragma (2026). *Two reads of the retained field*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/scalar_response/CONTINUATION.md

[34] FloatingPragma (2026). *Scalar preparation, evolution and detection through source controls*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/source_scalar/README.md

[35] FloatingPragma (2026). *A finite source implementation of the two-site scalar instrument*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/source_scalar/DERIVATION.md

[36] FloatingPragma (2026). *Execute the native C/E discrimination experiment from source controls*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/source_discrimination/README.md

[37] FloatingPragma (2026). *Source implementation of the complete C/E discriminator*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/source_discrimination/DERIVATION.md

[38] FloatingPragma (2026). *Fixed observer predictions and complete-count comparison*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/predictions/README.md

[39] FloatingPragma (2026). *Exact record-instrument discrimination*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/predictions/DERIVATION.md

[40] FloatingPragma (2026). *N4 memory survives the observer's local tick clock*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/local_tick/README.md

[41] FloatingPragma (2026). *N4 occupation memory at observer-local ticks*. Protocol specification. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/local_tick/SPEC.md

[42] FloatingPragma (2026). *A common interacting observer process*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/README.md

[43] FloatingPragma (2026). *One finite interacting source: derivation and theorem attachments*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/DERIVATION.md

[44] FloatingPragma (2026). *The interacting feedback signal survives infinite occupation*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/INFINITE_HISTORY.md

[45] FloatingPragma (2026). *Occupation refinement of the interacting retained history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/OCCUPATION.md

[46] FloatingPragma (2026). *Occupation limit of the same interacting observer history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/OCCUPATION_REFINEMENT.md

[47] FloatingPragma (2026). *Infinite coordinate-sign readout*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/READOUT.md

[48] FloatingPragma (2026). *A smooth binary observer with a complete occupation error bound*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/SMOOTH_REFINEMENT.md

[49] FloatingPragma (2026). The finite executable observer specification. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/COMPOSITION.md

[50] FloatingPragma (2026). Complete-output certificate for the continued smooth history. https://github.com/FloatingPragma/observer-patch-holography/blob/901b337f7fe689bea22112f23e73600842ba6f7e/model/common_source/SMOOTH_HISTORY.md

[51] FloatingPragma (2026). Coherent native repairs around one continuing quantum history. https://github.com/FloatingPragma/observer-patch-holography/blob/a59cc5e5640a7d34ee93a8a122241964074acac5/model/common_source/QUANTUM_NATIVE.md

[52] FloatingPragma (2026). Active encoded memory in the continuing observer. https://github.com/FloatingPragma/observer-patch-holography/blob/a59cc5e5640a7d34ee93a8a122241964074acac5/model/common_source/ENCODED_MEMORY.md

[53] FloatingPragma (2026). *Hilbert space, modular clocks and operational regions*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/OBSERVER_READOUTS.md

[54] FloatingPragma (2026). *A fixed thermal reference for the changing common history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/THERMAL_CLOCK.md

[55] FloatingPragma (2026). *One complete history from local constraint repair*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/HISTORY_REPAIR.md

[56] FloatingPragma (2026). *The existing three-dimensional response carrier on actual observer records*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/SLOW_GEOMETRY.md

[57] FloatingPragma (2026). *Exact native causal cones*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/CAUSALITY.md

[58] FloatingPragma (2026). *Field relations of the existing native repair*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/NATIVE_FIELDS.md

[59] FloatingPragma (2026). *Testing the readouts before assigning physical units*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/CALIBRATION.md

[60] FloatingPragma (2026). Retained receipt of the exact native observer histories and calculation checks. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/results/native.json

[61] FloatingPragma (2026). Retained receipt of the 20,000-trial sequential native execution. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/results/run.json

[62] FloatingPragma (2026). Retained receipt of the record capacity and compact controller. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/results/capacity.json

[63] FloatingPragma (2026). Retained receipt of the quantum mixing and memory certificates. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/results/quantum_mixing.json

[64] FloatingPragma (2026). Retained receipt of complete-record recovery and identifying schedules. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/results/record_recovery.json

[65] FloatingPragma (2026). Retained receipt of the P-dependent detector. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/p_response/results.json

[66] FloatingPragma (2026). Retained receipt of the local-tick memory certificate. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/local_tick/results.json

[67] FloatingPragma (2026). Retained receipt of the scalar detector on four grids. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/scalar_response/results.json

[68] FloatingPragma (2026). Retained receipt of the two-site source apparatus. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/observer/source_scalar/results.json

[69] FloatingPragma (2026). Retained receipt of the common interacting source. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/results.json

[70] FloatingPragma (2026). Retained receipt of the complete sharp-output comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/sharp_output_results.json

[71] FloatingPragma (2026). Retained receipt of the observer readouts and the modular clock. https://github.com/FloatingPragma/observer-patch-holography/blob/196b271021967927bc2626283b6cf7955972a4b2/model/common_source/bridge_results.json

[72] Aharonov, Y., and Bohm, D. (1959). "Significance of electromagnetic potentials in the quantum theory." *Physical Review* **115**(3), 485–491. https://doi.org/10.1103/PhysRev.115.485

[73] Bratteli, O., and Robinson, D. W. (1987). *Operator Algebras and Quantum Statistical Mechanics 1: C^*- and W^*-Algebras, Symmetry Groups, Decomposition of States*, 2nd ed. Springer, Berlin.

[74] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. Springer, Berlin.

[75] Caputo, P., Liggett, T. M., and Richthammer, T. (2010). "Proof of Aldous' spectral gap conjecture." *Journal of the American Mathematical Society* **23**(3), 831–851. https://arxiv.org/abs/0906.1238

[76] Choi, M.-D. (1974). "A Schwarz inequality for positive linear maps on C^*-algebras." *Illinois Journal of Mathematics* **18**(4), 565–574.

[77] Mohr, P. J., Newell, D. B., Taylor, B. N., and Tiesinga, E. (2025). "CODATA recommended values of the fundamental physical constants: 2022." *Reviews of Modern Physics* **97**, 025002. https://doi.org/10.1103/RevModPhys.97.025002

[78] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[79] Olver, F. W. J., et al., eds. *NIST Digital Library of Mathematical Functions*, Chapter 18, Orthogonal Polynomials. https://dlmf.nist.gov/18

[80] Gurvits, L., and Barnum, H. (2002). "Largest separable balls around the maximally mixed bipartite quantum state." *Physical Review A* **66**, 062311. https://doi.org/10.1103/PhysRevA.66.062311

[81] Hoeffding, W. (1963). "Probability inequalities for sums of bounded random variables." *Journal of the American Statistical Association* **58**(301), 13–30. https://doi.org/10.1080/01621459.1963.10500830

[82] Horodecki, M., Shor, P. W., and Ruskai, M. B. (2003). "Entanglement breaking channels." *Reviews in Mathematical Physics* **15**(6), 629–641. https://arxiv.org/abs/quant-ph/0302031

[83] Knill, E., and Laflamme, R. (1997). "Theory of quantum error-correcting codes." *Physical Review A* **55**, 900–911. https://doi.org/10.1103/PhysRevA.55.900

[84] Kraus, K. (1983). *States, Effects, and Operations: Fundamental Notions of Quantum Theory*. Lecture Notes in Physics 190, Springer, Berlin.

[85] Lüders, G. (1950). "Über die Zustandsänderung durch den Meßprozeß." *Annalen der Physik* **443**, 322–328. https://doi.org/10.1002/andp.19504430510

[86] Peres, A. (1996). "Separability criterion for density matrices." *Physical Review Letters* **77**, 1413–1415. https://doi.org/10.1103/PhysRevLett.77.1413

[87] Reed, M., and Simon, B. (1980). *Methods of Modern Mathematical Physics I: Functional Analysis*, revised and enlarged ed. Academic Press, New York.

[88] Reed, M., and Simon, B. (1975). *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*. Academic Press, New York.

[89] Strang, G. (1968). "On the construction and comparison of difference schemes." *SIAM Journal on Numerical Analysis* **5**(3), 506–517. https://doi.org/10.1137/0705041

[90] Teschl, G. (2014). *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, 2nd ed. Graduate Studies in Mathematics 157, American Mathematical Society, Providence.

[91] Acciaio, B., Beiglböck, M., Penkner, F., Schachermayer, W., and Temme, J. (2013). "A trajectorial interpretation of Doob's martingale inequalities." *The Annals of Applied Probability* **23**, 1494–1505. https://doi.org/10.1214/12-AAP878

[92] Andrews, B., and Clutterbuck, J. (2010). "Proof of the fundamental gap conjecture." arXiv:1006.1686. https://arxiv.org/abs/1006.1686

[93] Arias, A., Gheondea, A., and Gudder, S. (2002). "Fixed points of quantum operations." *Journal of Mathematical Physics* **43**, 5872–5881. https://doi.org/10.1063/1.1519669

[94] Azuma, K. (1967). "Weighted sums of certain dependent random variables." *Tohoku Mathematical Journal* **19**, 357–367. https://doi.org/10.2748/tmj/1178243286

[95] Banach, S. (1922). "Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales." *Fundamenta Mathematicae* **3**, 133–181. https://doi.org/10.4064/fm-3-1-133-181

[96] Brier, G. W. (1950). "Verification of forecasts expressed in terms of probability." *Monthly Weather Review* **78**, 1–3. https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

[97] Cavalletti, F., and Mondino, A. (2018). "Isoperimetric inequalities for finite perimeter sets under lower Ricci curvature bounds." *Rendiconti Lincei - Matematica e Applicazioni* **29**, 413–430. https://doi.org/10.4171/RLM/814

[98] Childs, A. M., Su, Y., Tran, M. C., Wiebe, N., and Zhu, S. (2021). "Theory of Trotter Error with Commutator Scaling." *Physical Review X* **11**, 011020. https://doi.org/10.1103/PhysRevX.11.011020

[99] Choi, M. D. (1975). "Completely positive linear maps on complex matrices." *Linear Algebra and its Applications* **10**, 285–290. https://doi.org/10.1016/0024-3795(75)90075-0

[100] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). "Proposed experiment to test local hidden-variable theories." *Physical Review Letters* **23**, 880–884. https://doi.org/10.1103/PhysRevLett.23.880

[101] Combescure, M., and Robert, D. (2012). *Coherent States and Applications in Mathematical Physics*. Springer. https://doi.org/10.1007/978-94-007-0196-0

[102] Coulhon, T., and Sikora, A. (2006). "Gaussian heat kernel upper bounds via Phragmén-Lindelöf theorem." *arXiv preprint arXiv:math/0609429*. https://doi.org/10.1112/plms/pdm050

[103] Dobrushin, R. L. (1956). "Central limit theorem for nonstationary Markov chains. I." *Theory of Probability & Its Applications* **1**, 65–80. https://doi.org/10.1137/1101006

[104] Feller, W. (1968). *An Introduction to Probability Theory and Its Applications, Volume 1*. Wiley. ISBN 0-471-25708-7.

[105] Girouard, A., and Polterovich, I. (2014). "Spectral geometry of the Steklov problem." *arXiv preprint arXiv:1411.6567*. https://doi.org/10.48550/arXiv.1411.6567

[106] Glauber, R. J. (1963). "Coherent and incoherent states of the radiation field." *Physical Review* **131**, 2766–2788. https://doi.org/10.1103/PhysRev.131.2766

[107] Gorini, V., Kossakowski, A., and Sudarshan, E. C. G. (1976). "Completely positive dynamical semigroups of N-level systems." *Journal of Mathematical Physics* **17**, 821–825. https://doi.org/10.1063/1.522979

[108] Gronwall, T. H. (1919). "Note on the derivatives with respect to a parameter of the solutions of a system of differential equations." *Annals of Mathematics* **20**, 292. https://doi.org/10.2307/1967124

[109] Haag, R., Hugenholtz, N. M., and Winnink, M. (1967). "On the equilibrium states in quantum statistical mechanics." *Communications in Mathematical Physics* **5**, 215–236. https://doi.org/10.1007/BF01646342

[110] Hensen, B., and others (2015). "Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres." *Nature* **526**, 682–686. https://doi.org/10.1038/nature15759

[111] Higuchi, Y., Segawa, E., and Suzuki, A. (2016). "Spectral mapping theorem of an abstract quantum walk." arXiv preprint, version 2. arXiv:1506.06457. https://arxiv.org/abs/1506.06457

[112] Holevo, A. (2011). *Probabilistic and Statistical Aspects of Quantum Theory*. Edizioni della Normale, Pisa. https://doi.org/10.1007/978-88-7642-378-9

[113] Horn, R. A., and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press. https://doi.org/10.1017/CBO9781139020411

[114] Jahnke, T., and Lubich, C. (2000). "Error Bounds for Exponential Operator Splittings." *BIT Numerical Mathematics* **40**, 735–744. https://doi.org/10.1023/A:1022396519656

[115] Jaksch, D., Bruder, C., Cirac, J. I., Gardiner, C. W., and Zoller, P. (1998). "Cold bosonic atoms in optical lattices." *Physical Review Letters* **81**, 3108. https://doi.org/10.1103/PhysRevLett.81.3108

[116] Jordan, P., and Wigner, E. (1928). "Über das Paulische Äquivalenzverbot." *Zeitschrift für Physik* **47**, 631–651. https://doi.org/10.1007/BF01331938

[117] Jordan, S. P., Lee, K. S. M., and Preskill, J. (2014). "Quantum Computation of Scattering in Scalar Quantum Field Theories." *Quantum Information and Computation* **14**, 1014–1080. arXiv:1112.4833. https://arxiv.org/abs/1112.4833

[118] Kahn, A. B. (1962). "Topological sorting of large networks." *Communications of the ACM* **5**, 558–562. https://doi.org/10.1145/368996.369025

[119] Kato, T. (1995). *Perturbation Theory for Linear Operators*. Springer, Berlin (Classics in Mathematics). https://doi.org/10.1007/978-3-642-66282-9

[120] Kemeny, J. G., and Snell, J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics). ISBN 0-387-90192-2.

[121] Kogut, J., and Susskind, L. (1975). "Hamiltonian formulation of Wilson's lattice gauge theories." *Physical Review D* **11**, 395–408. https://doi.org/10.1103/PhysRevD.11.395

[122] Kraus, K. (1971). "General state changes in quantum theory." *Annals of Physics* **64**, 311–335. https://doi.org/10.1016/0003-4916(71)90108-4

[123] Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*. American Mathematical Society. https://doi.org/10.1090/mbk/107

[124] Li, X., and Ding, P. (2016). "General forms of finite population central limit theorems with applications to causal inference." *arXiv preprint arXiv:1610.04821*. https://doi.org/10.48550/arXiv.1610.04821

[125] Lindblad, G. (1975). "Completely positive maps and entropy inequalities." *Communications in Mathematical Physics* **40**, 147–151. https://doi.org/10.1007/BF01609396

[126] Lindblad, G. (1976). "On the generators of quantum dynamical semigroups." *Communications in Mathematical Physics* **48**, 119–130. https://doi.org/10.1007/BF01608499

[127] Lovász, L. (1979). "On the Shannon capacity of a graph." *IEEE Transactions on Information Theory* **25**, 1–7. https://doi.org/10.1109/TIT.1979.1055985

[128] McGarry, C., Chalermpusitarak, T., Schwennicke, K., Scuccimarra, F., Millican, M. J., Matsos, V. G., et al. (2026). "Programmable quantum simulation of anharmonic dynamics." arXiv:2603.04744v1. https://arxiv.org/abs/2603.04744v1

[129] Moore, R. E., Kearfott, R. B., and Cloud, M. J. (2009). *Introduction to Interval Analysis*. SIAM. https://doi.org/10.1137/1.9780898717716

[130] Murphy, A. H. (1973). "A new vector partition of the probability score." *Journal of Applied Meteorology* **12**, 595–600. https://doi.org/10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2

[131] Nelson, E. (1959). "Analytic vectors." *Annals of Mathematics* **70**, 572. https://doi.org/10.2307/1970331

[132] Nielsen, M. A., and Chuang, I. L. (2010). *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press. https://doi.org/10.1017/CBO9780511976667

[133] Norris, J. R. (1997). *Markov Chains*. Cambridge University Press. https://doi.org/10.1017/CBO9780511810633

[134] Olver, F. W. J., Lozier, D. W., Boisvert, R. F., and Clark, C. W. (2010). *NIST Handbook of Mathematical Functions*. Cambridge University Press. ISBN 978-0-521-19225-5.

[135] Ozawa, M. (1984). "Quantum measuring processes of continuous observables." *Journal of Mathematical Physics* **25**, 79–87. https://doi.org/10.1063/1.526000

[136] Paulsen, V. (2002). *Completely Bounded Maps and Operator Algebras*. Cambridge University Press. https://doi.org/10.1017/CBO9780511546631

[137] Pazy, A. (1983). *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer. https://doi.org/10.1007/978-1-4612-5561-1

[138] Poulin, D. (2010). "Lieb-Robinson Bound and Locality for General Markovian Quantum Dynamics." *Physical Review Letters* **104**, 190401. https://doi.org/10.1103/PhysRevLett.104.190401

[139] Ramsey, N. F. (1950). "A Molecular Beam Resonance Method with Separated Oscillating Fields." *Physical Review* **78**, 695–699. https://doi.org/10.1103/PhysRev.78.695

[140] Reed, M., and Simon, B. (1972). "Unbounded operators." In *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, 249–317. Academic Press. https://doi.org/10.1016/B978-0-12-585001-8.50014-3

[141] Rivera, A. (2018). "Weighted local Weyl laws for elliptic operators." *arXiv preprint arXiv:1801.07598*. https://doi.org/10.48550/arXiv.1801.07598

[142] Rump, S. M. (2010). "Verification methods: rigorous results using floating-point arithmetic." *Acta Numerica* **19**, 287–449. https://doi.org/10.1017/S096249291000005X

[143] Ruskai, M. B. (1994). "Beyond strong subadditivity? Improved bounds on the contraction of generalized relative entropy." *Reviews in Mathematical Physics* **6**, 1147–1161. https://doi.org/10.1142/S0129055X94000407

[144] Shannon, C. (1956). "The zero error capacity of a noisy channel." *IRE Transactions on Information Theory* **2**, 8–19. https://doi.org/10.1109/TIT.1956.1056798

[145] Sheffield, S. (2007). "Gaussian free fields for mathematicians." *Probability Theory and Related Fields* **139**, 521–541. https://doi.org/10.1007/s00440-006-0050-1

[146] Stinespring, W. F. (1955). "Positive functions on C*-algebras." *Proceedings of the American Mathematical Society* **6**, 211–216. https://doi.org/10.1090/S0002-9939-1955-0069403-4

[147] Stoer, J., and Bulirsch, R. (2002). *Introduction to Numerical Analysis*. Springer, New York (Texts in Applied Mathematics). https://doi.org/10.1007/978-0-387-21738-3

[148] Szegedy, M. (2004). "Spectra of Quantized Walks and a sqrt(delta epsilon) rule." arXiv preprint. arXiv:quant-ph/0401053. https://arxiv.org/abs/quant-ph/0401053

[149] Szegő, G. (1939). *Orthogonal Polynomials*. American Mathematical Society (Colloquium Publications 23). https://doi.org/10.1090/coll/023

[150] Takesaki, M. (1970). *Tomita's Theory of Modular Hilbert Algebras and its Applications*. Springer. https://doi.org/10.1007/BFb0065832

[151] Tomiyama, J. (1957). "On the projection of norm one in W*-algebras." *Proceedings of the Japan Academy* **33**, 608–612. https://doi.org/10.3792/pja/1195524885

[152] Umegaki, H. (1954). "Conditional expectation in an operator algebra, I." *Tohoku Mathematical Journal* **6**, 177–181. https://doi.org/10.2748/tmj/1178245177

[153] Watrous, J. (2018). *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142

[154] Weedbrook, C., Pirandola, S., García-Patrón, R., Cerf, N. J., Ralph, T. C., Shapiro, J. H., et al. (2012). "Gaussian quantum information." *Reviews of Modern Physics* **84**, 621–669. https://doi.org/10.1103/RevModPhys.84.621

[155] Welch, J., Greenbaum, D., Mostame, S., and Aspuru-Guzik, A. (2014). "Efficient quantum circuits for diagonal unitaries without ancillas." *New Journal of Physics* **16**, 033040. https://doi.org/10.1088/1367-2630/16/3/033040

[156] Whitt, W. (2007). "Proofs of the martingale FCLT." *Probability Surveys* **4**, 268–302. https://doi.org/10.1214/07-PS122

[157] Zhang, F. (2005). *The Schur Complement and Its Applications*. Springer. https://doi.org/10.1007/b105056

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.